

GENERAL APTITUDE

Previous Year Questions • 2015–2020

GATE — ALL BRANCHES

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in General Aptitude (all GATE branches), gathered from the original papers and arranged year by year and session by session. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by year and by session, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q24.5.3

24	GATE exam year — 2024 (two digits; 98 = 1998)
5	Session number of that year's exam
3	Question number within that session

So **Q24.5.3** is the 3rd aptitude question of Session 5, GATE 2024. Numbers are therefore not sequential through the book — they tell you the year and session at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a session to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

  Easy  Medium  Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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Contents

Year	Sessions	Pages
2020		**1**
	Session : 1	2
	Session : 2	3
	Session : 3	4
	Session : 4	6
	Session : 5	7
	Session : 6	8
	Session : 7	10
	Session : 8	11
2019		**14**
	Session : 1	15
	Session : 2	16
	Session : 3	17
	Session : 4	19
	Session : 5	20
	Session : 6	22
	Session : 7	23
	Session : 8	24
2018		**27**
	Session : 1	28
	Session : 2	29
	Session : 3	30
	Session : 4	31
	Session : 5	32
	Session : 6	33
	Session : 7	34
	Session : 8	35
2017		**38**
	Session : 1	39
	Session : 2	40
	Session : 3	41
	Session : 4	42
	Session : 5	44

Session : 6 45
Session : 7 46
Session : 8 47

2016 **50**
Session : 1 51
Session : 2 52
Session : 3 53
Session : 4 54
Session : 5 55
Session : 6 56
Session : 7 58
Session : 8 59

2015 **62**
Session : 1 63
Session : 2 64
Session : 3 65
Session : 4 66
Session : 5 67
Session : 6 69
Session : 7 70
Session : 8 71

YEAR

2020

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	IN, ME1, MT, PE, PH	10
Session 2	CY, ME2, PI	10
Session 3	AR, BM, BT, CH, MA, MN, ST, XE, XL	10
Session 4	AE, AG, EC, GG	10
Session 5	EE, EY, TF	10
Session 6	CS	10
Session 7	CE1	10
Session 8	CE2	10

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Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 01/02/20

Q20.1.1 MCQ 2020 1M

He is known for his unscrupulous ways. He always sheds _____ tears to deceive people.

A. fox's **B.** crocodile's **C.** crocodile **D.** fox

Q20.1.2 MCQ 2020 1M

Jofra Archer, the England fast bowler, is _____ than accurate.

A. more fast **B.** faster **C.** less fast **D.** more faster

Q20.1.3 MCQ 2020 1M

Select the word that fits the analogy:
Build : Building :: Grow : _____

A. Grown **B.** Grew **C.** Growth **D.** Growed

Q20.1.4 MCQ 2020 1M

I do not think you know the case well enough to have opinions. Having said that, I agree with your other point. What does the phrase "having said that" mean in the given text?

A. as opposed to what I have said

B. despite what I have said

C. in addition to what I have said

D. contrary to what I have said

Q20.1.5 MCQ 2020 1M

Define $[x]$ as the greatest integer less than or equal to x, for each $x \in (-\infty, \infty)$. If $y = [x]$, then area under y for $x \in [1, 4]$ is _____.

A. 1 **B.** 3 **C.** 4 **D.** 6

Q20.1.6 MCQ 2020 2M

Crowd funding deals with mobilisation of funds for a project from a large number of people, who would be willing to invest smaller amounts through web-based platforms in the project.
Based on the above paragraph, which of the following is correct about crowd funding?

A. Funds raised through unwilling contributions on web-based platforms.

B. Funds raised through large contributions on web-based platforms.

C. Funds raised through coerced contributions on web-based platforms.

D. Funds raised through voluntary contributions on web-based platforms.

Q20.1.7 MCQ 2020 2M

P, Q, R and S are to be uniquely coded using α and β. If P is coded as $\alpha\alpha$ and Q as $\alpha\beta$, then R and S, respectively, can be coded as _____.

A. $\beta\alpha$ and $\alpha\beta$ **B.** $\beta\beta$ and $\alpha\alpha$

C. $\alpha\beta$ and $\beta\beta$ **D.** $\beta\alpha$ and $\beta\beta$

Q20.1.8 MCQ 2020 2M

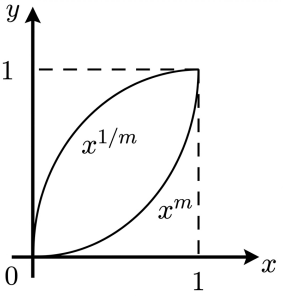
The sum of the first n terms in the sequence $8, 88, 888, 8888, \ldots$ is _____.

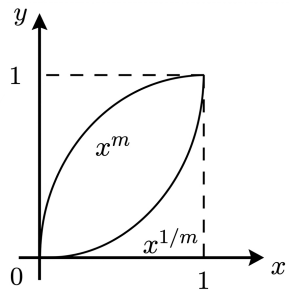
A. $\frac{81}{80}(10^n - 1) + \frac{9}{8}n$ **B.** $\frac{81}{80}(10^n - 1) - \frac{9}{8}n$

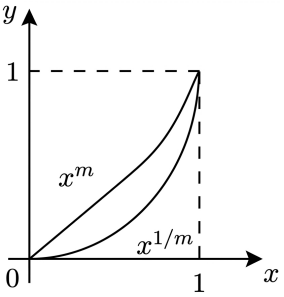
C. $\frac{80}{81}(10^n - 1) + \frac{8}{9}n$ **D.** $\frac{80}{81}(10^n - 1) - \frac{8}{9}n$

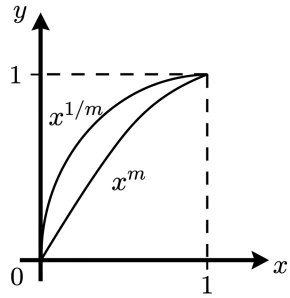
Q20.1.9 MCQ 2020 2M

Select the graph that schematically represents BOTH $y = x^m$ and $y = x^{1/m}$ properly in the interval $0 \le x \le 1$, for integer values of m, where $m > 1$.

A.


B.


C.


D.


Q20.1.10 MCQ 2020 2M

The bar graph shows the data of the students who appeared and passed in an examination for four schools P, Q, R and S. The average of success rates (in percentage) of these four schools is _____.

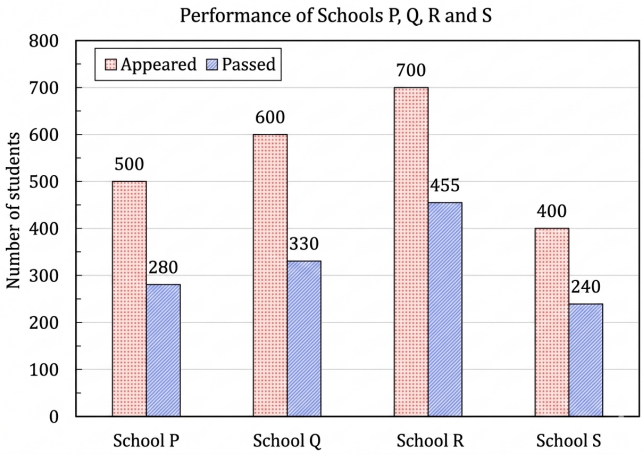


Fig. Q20.1.10

A. 58.5 % **B.** 58.8 % **C.** 59.0 % **D.** 59.3 %

Session : 2 | Afternoon | 01/02/20

Q20.2.1 MCQ 2020 1M

While I agree _____ his proposal this time, I do not often agree _____ him.

A. to, with **B.** with, to **C.** with, with **D.** to, to

Q20.2.2 MCQ 2020 1M

The recent measures to improve the output would _____ the level of production to our satisfaction.

A. increase **B.** decrease **C.** speed **D.** equalise

Q20.2.3 MCQ 2020 1M

Select the word that fits the analogy:
White: Whitening : : Light: _____

A. Lightning **B.** Lightening

C. Lighting **D.** Enlightening

Q20.2.4 MCQ 2020 1M

In one of the greatest innings ever seen in 142 years of Test history, Ben Stokes upped the tempo in a five-and-a-half hour long stay of 219 balls including 11 fours and 8 sixes that saw him finish on a 135 not out as England squared the five-match series.
Based on their connotations in the given passage, which one of the following meanings DOES NOT match?

A. upped = increased **B.** squared = lost

C. tempo = enthusiasm **D.** saw = resulted in

Q20.2.5 MCQ 2020 1M

There are five levels $\{P, Q, R, S, T\}$ in a linear supply chain before a product reaches customers, as shown in the figure.

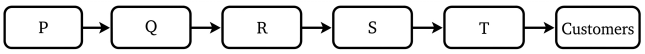


Fig. Q20.2.5

At each of the five levels, the price of the product is increased by 25%. If the product is produced at level P at the cost of Rs. 120 per unit, what is the price paid (in rupees) by the customers?

A. 187.50 **B.** 234.38 **C.** 292.96 **D.** 366.21

Q20.2.6 MCQ 2020 2M

Climate change and resilience deal with two aspects – reduction of sources of non-renewable energy resources and reducing vulnerability of climate change aspects. The terms 'mitigation' and 'adaptation' are used to refer to these aspects, respectively.
Which of the following assertions is best supported by the above information?

A. Mitigation deals with consequences of climate change

B. Adaptation deals with causes of climate change

C. Mitigation deals with actions taken to reduce the use of fossil fuels

D. Adaptation deals with actions taken to combat greenhouse gas emissions

Q20.2.7 MCQ 2020 2M

Find the missing element in the following figure.

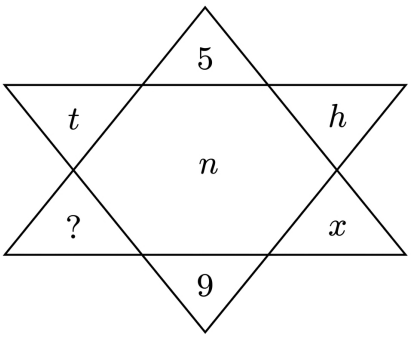


Fig. Q20.2.7

A. d **B.** e **C.** w **D.** y

Q20.2.8 MCQ 2020 2M

It was estimated that 52 men can complete a strip in a newly constructed highway connecting cities P and Q in 10 days. Due to an emergency, 12 men were sent to another project. How many number of days, more than the original estimate, will be required to complete the strip?

A. 3 days **B.** 5 days

C. 10 days **D.** 13 days

Q20.2.9 MCQ 2020 2M

An engineer measures THREE quantities X, Y and Z in an experiment. She finds that they follow a relationship that is represented in the figure below: (the product of X and Y linearly varies with Z)

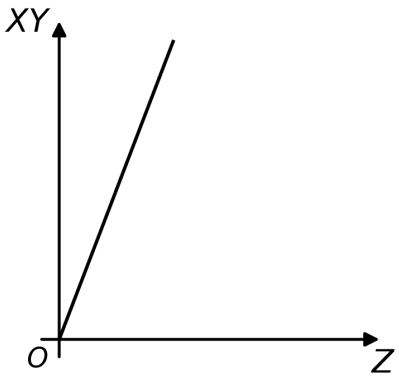


Fig. Q20.2.9

Then, which of the following statements is FALSE?

A. For fixed Z; X is proportional to Y

B. For fixed Y; X is proportional to Z

C. For fixed X; Z is proportional to Y

D. XY/Z is constant

Q20.2.10 MCQ 2020 2M

The two pie-charts given below show the data of total students and only girls registered in different streams in a university. If the total number of students registered in the university is 5000, and the total number of the registered girls is 1500; then, the ratio of boys enrolled in Arts to the girls enrolled in Management is _____.

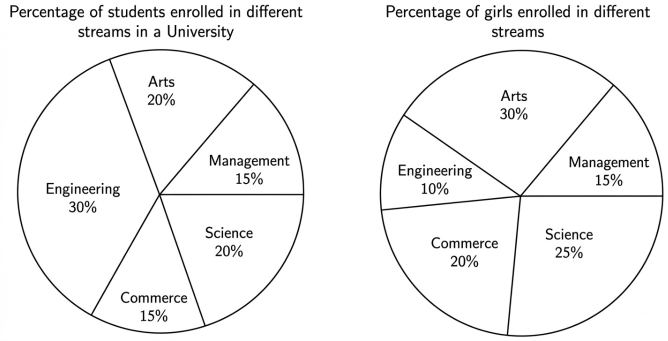


Fig. Q20.2.10

A. 2 : 1 **B.** 9 : 22 **C.** 11 : 9 **D.** 22 : 9

Session : 3 | Forenoon | 02/02/20

Q20.3.1 MCQ 2020 1M

Rajiv Gandhi Khel Ratna Award was conferred ____ Mary Kom, a six-time world champion in boxing, recently in a ceremony ____ the Rashtrapati Bhawan (the President's official residence) in New Delhi.

A. with, at **B.** on, in **C.** on, at **D.** to, at

Q20.3.2 MCQ 2020 1M

Despite a string of poor performances, the chances of K. L. Rahul's selection in the team are _____.

A. slim **B.** bright **C.** obvious **D.** uncertain

Q20.3.3 MCQ 2020 1M

Select the word that fits the analogy:
Cover : Uncover :: Associate : _____

A. Unassociate **B.** Inassociate

C. Misassociate **D.** Dissociate

Q20.3.4 MCQ 2020 1M

Hit by floods, the Kharif (summer sown) crops in various parts of the country have been affected. Officials believe that the loss in production of the kharif crops can be recovered in the output of the rabi (winter sown) crops so that the country can achieve its food-grain production target of 291 million tons in the crop year 2019-20 (July-June). They are hopeful that good rains in July-August will help the soil retain moisture for a longer period, helping winter sown crops such as wheat and pulses during the November-February period.
Which of the following statements can be inferred from the given passage?

A. Officials want the food-grain production target to be met by the November-February period.

B. Officials feel that the food-grain production target cannot be met due to floods.

C. Officials declared that the food-grain production target will be met due to good rains.

D. Officials hope that the food-grain production target will be met due to good rabi produce.

Q20.3.5 MCQ | 2020 | 1M

The difference between the sum of the first $2n$ natural numbers and the sum of the first n odd natural numbers is ___.

A. $n^2 - n$ **B.** $n^2 + n$ **C.** $2n^2 - n$ **D.** $2n^2 + n$

Q20.3.6 MCQ | 2020 | 2M

Repo rate is the rate at which Reserve Bank of India (RBI) lends commercial banks, and reverse repo rate is the rate at which RBI borrows money from commercial banks.
Which of the following statements can be inferred from the above passage?

A. Decrease in repo rate will increase cost of borrowing and decrease lending by commercial banks.

B. Increase in repo rate will decrease cost of borrowing and increase lending by commercial banks.

C. Increase in repo rate will decrease cost of borrowing and decrease lending by commercial banks.

D. Decrease in repo rate will decrease cost of borrowing and increase lending by commercial banks.

Q20.3.7 MCQ | 2020 | 2M

P, Q, R, S, T, U, V, and W are seated around a circular table.

I. S is seated opposite to W.
II. U is seated at the second place to the right of R.
III. T is seated at the third place to the left of R.
IV. V is a neighbour of S.

Which of the following must be true?

A. P is a neighbour of R.

B. Q is a neighbour of R.

C. P is not seated opposite to Q.

D. R is the left neighbour of S.

Q20.3.8 MCQ | 2020 | 2M

The distance between Delhi and Agra is 233 km. A car P started travelling from Delhi to Agra and another car Q started from Agra to Delhi along the same road 1 hour after the car P started. The two cars crossed each other 75 minutes after the car Q started. Both cars were travelling at constant speed. The speed of car P was 10 km/hr more than the speed of car Q. How many kilometers the car Q had travelled when the cars crossed each other?

A. 66.6 **B.** 75.2 **C.** 88.2 **D.** 116.5

Q20.3.9 MCQ | 2020 | 2M

For a matrix $M = [m_{ij}]; i, j = 1, 2, 3, 4$, the diagonal elements are all zero and $m_{ij} = -m_{ji}$.
The minimum number of elements required to fully specify the matrix is ____.

A. 0 **B.** 6 **C.** 12 **D.** 16

Q20.3.10 MCQ | 2020 | 2M

The profit shares of two companies P and Q are shown in the figure. If the two companies have invested a fixed and equal amount every year, then the ratio of the total revenue of company P to the total revenue of company Q, during 2013 - 2018 is _____.

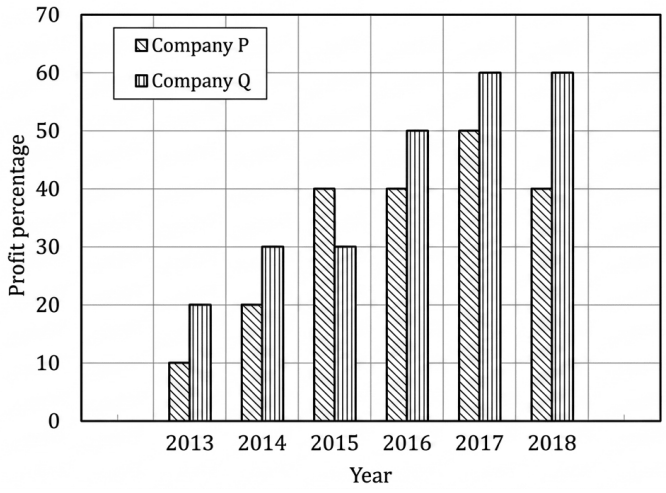


Fig. Q20.3.10

A. 15 : 17 **B.** 16 : 17 **C.** 17 : 15 **D.** 17 : 16

Session : 4 | Afternoon | 02/02/20

Q20.4.1 MCQ 2020 1M

The untimely loss of life is a cause of serious global concern as thousands of people get killed _____ accidents every year while many other die _____ diseases like cardio vascular disease, cancer, etc.

A. in, of **B.** from, of **C.** during, from **D.** from, from

Q20.4.2 MCQ 2020 1M

He was not only accused of theft _____ of conspiracy.

A. rather **B.** but also **C.** but even **D.** rather than

Q20.4.3 MCQ 2020 1M

Select the word that fits the analogy:
Explicit: Implicit :: Express: _____

A. Impress **B.** Repress **C.** Compress **D.** Suppress

Q20.4.4 MCQ 2020 1M

The Canadian constitution requires that equal importance be given to English and French. Last year, Air Canada lost a lawsuit, and had to pay a six-figure fine to a French-speaking couple after they filed complaints about formal in-flight announcements in English lasting 15 seconds, as opposed to informal 5 second messages in French.
The French-speaking couple were upset at _____.

A. the in-flight announcements being made in English.

B. the English announcements being clearer than the French ones.

C. the English announcements being longer than the French ones.

D. equal importance being given to English and French.

Q20.4.5 MCQ 2020 1M

A superadditive function $f(\cdot)$ satisfies the following property

$$f(x_1 + x_2) \geq f(x_1) + f(x_2)$$

Which of the following functions is a superadditive function for $x > 1$?

A. e^x **B.** $\sqrt{x}$ **C.** $1/x$ **D.** e^{-x}

Q20.4.6 MCQ 2020 2M

The global financial crisis in 2008 is considered to be the most serious world-wide financial crisis, which started with the sub-prime lending crisis in USA in 2007. The sub-prime lending crisis led to the banking crisis in 2008 with the collapse of Lehman Brothers in 2008. The sub-prime lending refers to the provision of loans to those borrowers who may have difficulties in repaying loans, and it arises because of excess liquidity following the East Asian crisis. Which one of the following sequences shows the correct precedence as per the given passage?

A. East Asian crisis→ subprime lending crisis → banking crisis→ global financial crisis.

B. Subprime lending crisis→ global financial crisis→ banking crisis→ East Asian crisis.

C. Banking crisis→ subprime lending crisis→ global financial crisis → East Asian crisis.

D. Global financial crisis → East Asian crisis → banking crisis → subprime lending crisis.

Q20.4.7 MCQ 2020 2M

It is quarter past three in your watch. The angle between the hour hand and the minute hand is _____.

A. 0° **B.** 7.5° **C.** 15° **D.** 22.5°

Q20.4.8 MCQ 2020 2M

A circle with centre O is shown in the figure. A rectangle PQRS of maximum possible area is inscribed in the circle. If the radius of the circle is a, then the area of the shaded portion is _____.

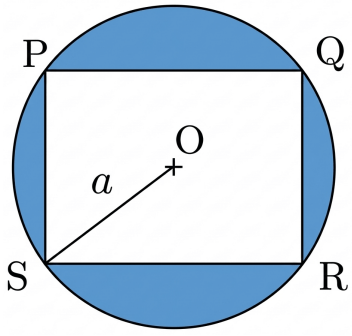


Fig. Q20.4.8

A. $\pi a^2 - a^2$ **B.** $\pi a^2 - \sqrt{2}a^2$

C. $\pi a^2 - 2a^2$ **D.** $\pi a^2 - 3a^2$

Q20.4.9 MCQ 2020 2M

a, b, c are real numbers. The quadratic equation $ax^2 - bx + c = 0$ has equal roots, which is β, then

A. $\beta = b/a$ **B.** $\beta^2 = ac$

C. $\beta^3 = bc/(2a^2)$ **D.** $b^2 \neq 4ac$

Q20.4.10 MCQ 2020 2M

The following figure shows the data of students enrolled in 5 years (2014 to 2018) for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is

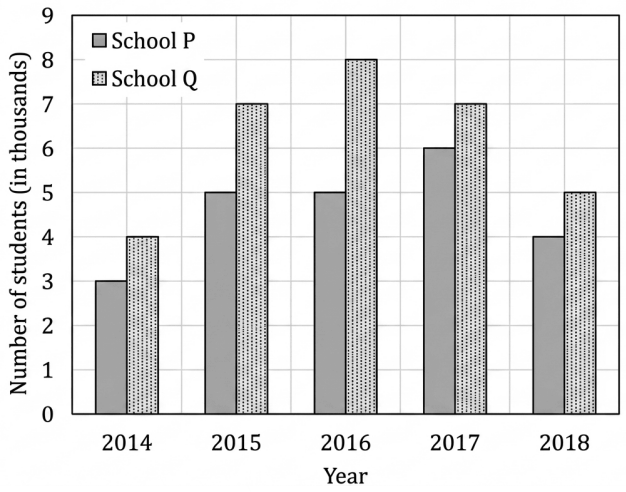


Fig. Q20.4.10

A. 8 : 23 **B.** 23 : 8 **C.** 23 : 31 **D.** 31 : 23

Session : 5 | Forenoon | 08/02/20

Q20.5.1 MCQ 2020 1M

This book, including all its chapters, ____ interesting. The students as well as the instructor ____ in agreement about it.

A. is, was **B.** are, are **C.** is, are **D.** were, was

Q20.5.2 MCQ 2020 1M

People were prohibited _____ their vehicles near the entrance of the main administrative building.

A. to park **B.** from parking

C. parking **D.** to have parked

Q20.5.3 MCQ 2020 1M

Select the word that fits the analogy:
Do : Undo :: Trust : _____

A. Entrust **B.** Intrust **C.** Distrust **D.** Untrust

Q20.5.4 MCQ 2020 1M

Stock markets _____ at the news of the coup.

A. poised **B.** plunged **C.** plugged **D.** probed

Q20.5.5 MCQ 2020 1M

If P, Q, R, S are four individuals, how many teams of size exceeding one can be formed, with Q as a member?

A. 5 **B.** 6 **C.** 7 **D.** 8

Q20.5.6 MCQ 2020 2M

Non-performing Assets (NPAs) of a bank in India is defined as an asset, which remains unpaid by a borrower for a certain period of time in terms of interest, principal, or both. Reserve Bank of India (RBI) has changed the definition of NPA thrice during 1993-2004, in terms of the holding period of loans. The holding period was reduced by one quarter each time. In 1993, the holding period was four quarters (360 days).
Based on the above paragraph, the holding period of loans in 2004 after the third revision was _____ days.

A. 45 **B.** 90 **C.** 135 **D.** 180

Q20.5.7 MCQ 2020 2M

Select the next element of the series: Z, WV, RQP, ____

A. LKJI **B.** JIHG **C.** KJIH **D.** NMLK

Q20.5.8 MCQ 2020 2M

In four-digit integer numbers from 1001 to 9999, the digit group "37" (in the same sequence) appears ____ times.

A. 270 **B.** 279 **C.** 280 **D.** 299

Q20.5.9 MCQ 2020 2M

Given a semicircle with O as the centre, as shown in the figure, the ratio $\frac{\overline{AC} + \overline{CB}}{\overline{AB}}$ is ____, where $\overline{AC}$, $\overline{CB}$ and $\overline{AB}$ are chords.

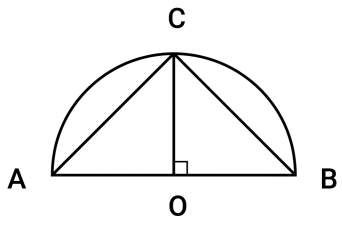


Fig. Q20.5.9

A. $\sqrt{2}$ **B.** $\sqrt{3}$ **C.** 2 **D.** 3

Q20.5.10 MCQ | 2020 | 2M

The revenue and expenditure of four different companies P, Q, R and S in 2015 are shown in the figure. If the revenue of company Q in 2015 was 20% more than that in 2014, and company Q had earned a profit of 10% on expenditure in 2014, then its expenditure (in million rupees) in 2014 was _____.

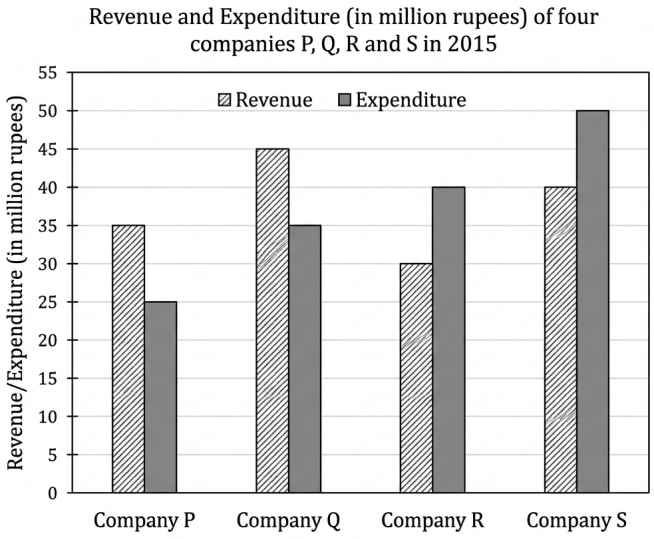


Fig. Q20.5.10

A. 32.7 **B.** 33.7 **C.** 34.1 **D.** 35.1

Session : 6 | Afternoon | 08/02/20

Q20.6.1 MCQ | 2020 | 1M

Raman is confident of speaking English _____ six months as he has been practising regularly _____ the last three weeks.

A. during, for **B.** for, since **C.** for, in **D.** within, for

Q20.6.2 MCQ | 2020 | 1M

His knowledge of the subject was excellent but his classroom performance was _____.

A. extremely poor **B.** good

C. desirable **D.** praiseworthy

Q20.6.3 MCQ | 2020 | 1M

Select the word that fits the analogy:
Cook : Cook :: Fly : _____

A. Flyer **B.** Flying **C.** Flew **D.** Flighter

Q20.6.4 MCQ | 2020 | 1M

The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones.
Which one of the following statements can be inferred from the given passage?

A. Nation-states do not have environmental concerns.

B. Nation-states are responsible for providing fresh water to billions of people.

C. Billions of people are responsible for man-made emissions.

D. Billions of people are affected by melting glaciers.

Q20.6.5 MCQ 2020 1M

There are multiple routes to reach from node 1 to node 2, as shown in the network.

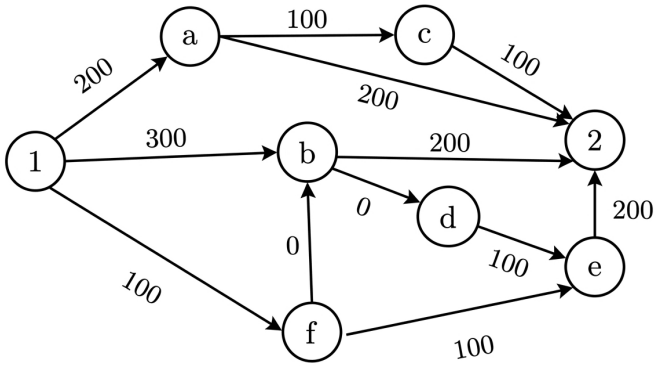


Fig. Q20.6.5

The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

A. 1-a-c-2 **B.** 1-f-b-2 **C.** 1-b-2 **D.** 1-f-e-2

Q20.6.6 MCQ 2020 2M

Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs). Which one of the following statements can be inferred from the given passage?

A. GST is imposed on the production of goods and services.

B. GST includes all indirect taxes.

C. GST does not have a component specific to UT.

D. GST is imposed at the point of usage of goods and services.

Q20.6.7 MCQ 2020 2M

If $P = 3$, $R = 27$, $T = 243$, then $Q + S =$ _____.

A. 40 **B.** 80 **C.** 90 **D.** 110

Q20.6.8 MCQ 2020 2M

The figure below shows an annular ring with outer and inner radii as b and a, respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is _____.

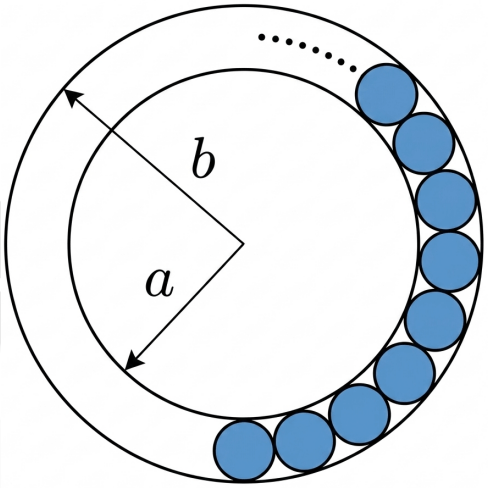


Fig. Q20.6.8

A. $\pi\left[(b^2 - a^2) - \frac{n}{4}(b-a)^2\right]$

B. $\pi\left[(b^2 - a^2) - n(b-a)^2\right]$

C. $\pi\left[(b^2 - a^2) + \frac{n}{4}(b-a)^2\right]$

D. $\pi\left[(b^2 - a^2) + n(b-a)^2\right]$

Q20.6.9 MCQ 2020 2M

Two straight lines are drawn perpendicular to each other in the X-Y plane. If α and β are the acute angles the straight lines make with the X-axis, then $\alpha + \beta$ is _____.

A. 60° **B.** 90° **C.** 120° **D.** 180°

Q20.6.10 MCQ 2020 2M

The total revenue of a company during 2014-2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014-2018 is _____.

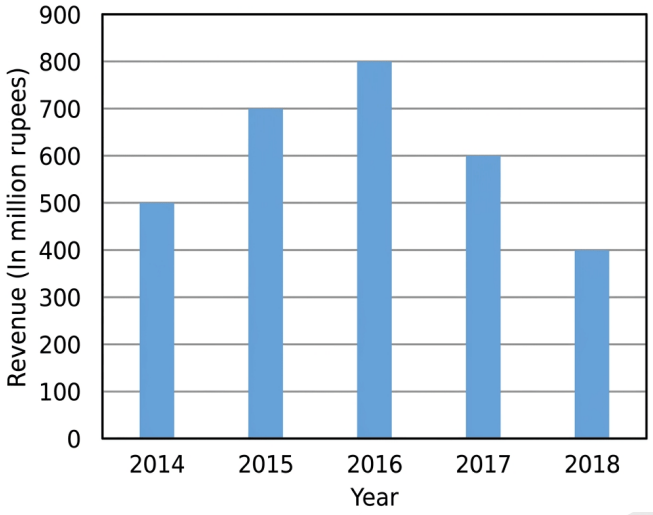


Fig. Q20.6.10

A. 16.67% profit **B.** 16.67% loss

C. 20% profit **D.** 20% loss

Session : 7 | Forenoon | 09/02/20

Q20.7.1 MCQ 2020 1M

It is a common criticism that most of the academicians live in their _____, so, they are not aware of the real life challenges.

A. homes **B.** ivory towers

C. glass palaces **D.** big flats

Q20.7.2 MCQ 2020 1M

His hunger for reading is insatiable. He reads indiscriminately. He is most certainly a/an _____ reader.

A. all-round **B.** precocious **C.** voracious **D.** wise

Q20.7.3 MCQ 2020 1M

Select the word that fits the analogy:
Fuse : Fusion :: Use : _____

A. Usage **B.** User **C.** Uses **D.** Usion

Q20.7.4 MCQ 2020 1M

If $0, 1, 2, \ldots, 7, 8, 9$ are coded as O, P, Q, . . . , V, W, X, then 45 will be coded as _____.

A. TS **B.** ST **C.** SS **D.** SU

Q20.7.5 MCQ 2020 1M

The sum of two positive numbers is 100. After subtracting 5 from each number, the product of the resulting numbers is 0. One of the original numbers is _____.

A. 80 **B.** 85 **C.** 90 **D.** 95

Q20.7.6 MCQ 2020 2M

The American psychologist Howard Gardner expounds that human intelligence can be sub-categorised into multiple kinds, in such a way that individuals differ with respect to their relative competence in each kind. Based on this theory, modern educationists insist on prescribing multi-dimensional curriculum and evaluation parameters that enable development and assessment of multiple intelligences. Which of the following statements can be inferred from the given text?

A. Howard Gardner insists that the teaching curriculum and evaluation needs to be multi-dimensional.

B. Howard Gardner wants to develop and assess the theory of multiple intelligences.

C. Modern educationists want to develop and assess the theory of multiple intelligences.

D. Modern educationists insist that the teaching curriculum and evaluation needs to be multi-dimensional.

Q20.7.7 MCQ 2020 2M

Five friends P, Q, R, S and T went camping. At night, they had to sleep in a row inside the tent. P, Q and T refused to sleep next to R since he snored loudly. P and S wanted to avoid Q as he usually hugged people in sleep.
Assuming everyone was satisfied with the sleeping arrangements, what is the order in which they slept?

A. RSPTQ **B.** SPRTQ **C.** QRSPT **D.** QTSPR

Q20.7.8 MCQ 2020 2M

Insert seven numbers between 2 and 34, such that the resulting sequence including 2 and 34 is an arithmetic progression. The sum of these inserted seven numbers is_____.

A. 120 **B.** 124 **C.** 126 **D.** 130

Q20.7.9 MCQ 2020 2M

The unit's place in 26591749^{110016} is _____.

A. 1 **B.** 3 **C.** 6 **D.** 9

Q20.7.10 MCQ 2020 2M

The total expenditure of a family, on different activities in a month, is shown in the pie-chart. The extra money spent on education as compared to transport (in percent) is _____.

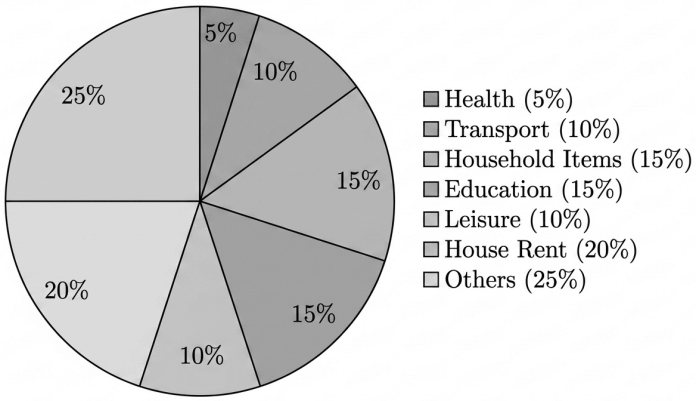


Fig. Q20.7.10

A. 5 **B.** 33.3 **C.** 50 **D.** 100

Session : 8 | Afternoon | 09/02/20

Q20.8.1 MCQ 2020 1M

Rescue teams deployed _____ disaster hit areas combat _____ a lot of difficulties to save the people.

A. with, at **B.** in, with **C.** with, with **D.** to, to

Q20.8.2 MCQ 2020 1M

Select the most appropriate word that can replace the underlined word without changing the meaning of the sentence.
Now-a-days, most children have a tendency to belittle the legitimate concerns of their parents.

A. disparage **B.** applaud **C.** reduce **D.** begrudge

Q20.8.3 MCQ 2020 1M

Select the word that fits the analogy:
Partial : Impartial :: Popular : _____

A. Impopular **B.** Dispopular

C. Mispopular **D.** Unpopular

Q20.8.4 MCQ 2020 1M

After the inauguration of the new building, the Head of the Department (HoD) collated faculty preferences for office space. *P* wanted a room adjacent to the lab. *Q* wanted to be close to the lift. *R* wanted a view of the playground and *S* wanted a corner office.
Assuming that everyone was satisfied, which among the following shows a possible allocation?

A.

	PLAYGROUND			
ROAD	HoD	Q	R	S
	P			LIFT
	LAB			

B.

	PLAYGROUND			
ROAD	S	R	P	HoD
	Q			LIFT
	LAB			

C.

	PLAYGROUND			
ROAD	S	R	HoD	Q
	P			LIFT
	LAB			

D.

	PLAYGROUND			
ROAD	HoD	S	R	Q
	P			LIFT
	LAB			

Q20.8.5 MCQ 2020 1M

If $f(x) = x^2$ for each $x \in (-\infty, \infty)$, then $\frac{f(f(f(x)))}{f(x)}$ is equal to _____.

A. $f(x)$ **B.** $(f(x))^2$ **C.** $(f(x))^3$ **D.** $(f(x))^4$

Q20.8.6 MCQ 2020 2M

Nominal interest rate is defined as the amount paid by the borrower to the lender for using the borrowed amount for a specific period of time. Real interest rate calculated on the basis of actual value (inflation-adjusted), is approximately equal to the difference between nominal rate and expected rate of inflation in the economy.
Which of the following assertions is best supported by the above information?

A. Under high inflation, real interest rate is low and borrowers get benefited.

B. Under low inflation, real interest rate is high and borrowers get benefited.

C. Under high inflation, real interest rate is low and lenders get benefited.

D. Under low inflation, real interest rate is low and borrowers get benefited.

Q20.8.7 MCQ 2020 2M

For the year 2019, which of the previous year's calendar can be used?

A. 2011 **B.** 2012 **C.** 2013 **D.** 2014

Q20.8.8 MCQ 2020 2M

The ratio of 'the sum of the odd positive integers from 1 to 100' to 'the sum of the even positive integers from 150 to 200' is _____.

A. 45 : 95 **B.** 1 : 2 **C.** 50 : 91 **D.** 1 : 1

Q20.8.9 MCQ 2020 2M

In a school of 1000 students, 300 students play chess and 600 students play football. If 50 students play both chess and football, the number of students who play neither is _____.

A. 200 **B.** 150 **C.** 100 **D.** 50

Q20.8.10 MCQ 2020 2M

The monthly distribution of 9 Watt LED bulbs sold by two firms X and Y from January to June 2018 is shown in the pie-chart and the corresponding table. If the total number of LED bulbs sold by two firms during April–June 2018 is 50000, then the number of LED bulbs sold by firm Y during April–June 2018 is _____.

Month	Ratio of LED bulbs sold by two firms (X : Y)
January	7 : 8
February	2 : 3
March	2 : 1
April	3 : 2
May	1 : 4
June	9 : 11

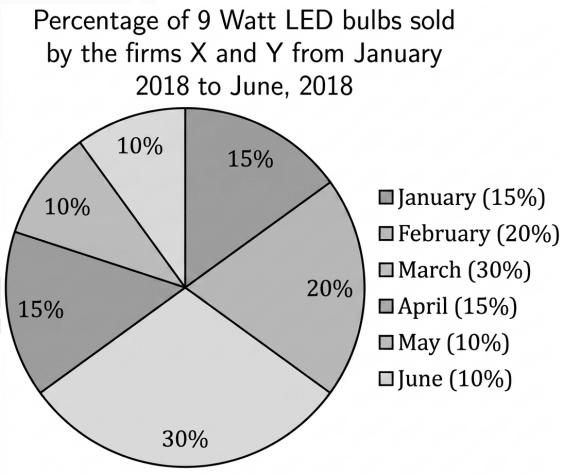


Fig. Q20.8.10

A. 11250 **B.** 9750 **C.** 8750 **D.** 8250

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q20.1.1 C	Q20.1.2 A	Q20.1.3 C	Q20.1.4 B	Q20.1.5 D	Q20.1.6 D
Q20.1.7 D	Q20.1.8 D	Q20.1.9 A	Q20.1.10 C		
Session : 2					
Q20.2.1 A	Q20.2.2 A	Q20.2.3 B	Q20.2.4 B	Q20.2.5 D	Q20.2.6 C
Q20.2.7 A	Q20.2.8 A	Q20.2.9 A	Q20.2.10 D		
Session : 3					
Q20.3.1 C	Q20.3.2 B	Q20.3.3 D	Q20.3.4 D	Q20.3.5 B	Q20.3.6 D
Q20.3.7 C	Q20.3.8 B	Q20.3.9 B	Q20.3.10 B		
Session : 4					
Q20.4.1 A	Q20.4.2 B	Q20.4.3 B	Q20.4.4 C	Q20.4.5 A	Q20.4.6 A
Q20.4.7 B	Q20.4.8 C	Q20.4.9 C	Q20.4.10 B		
Session : 5					
Q20.5.1 C	Q20.5.2 B	Q20.5.3 C	Q20.5.4 B	Q20.5.5 C	Q20.5.6 B
Q20.5.7 C	Q20.5.8 C	Q20.5.9 A	Q20.5.10 C		
Session : 6					
Q20.6.1 D	Q20.6.2 A	Q20.6.3 A	Q20.6.4 D	Q20.6.5 B	Q20.6.6 D
Q20.6.7 C	Q20.6.8 A	Q20.6.9 B	Q20.6.10 C		
Session : 7					
Q20.7.1 B	Q20.7.2 C	Q20.7.3 A	Q20.7.4 B	Q20.7.5 D	Q20.7.6 D
Q20.7.7 A	Q20.7.8 C	Q20.7.9 A	Q20.7.10 C		
Session : 8					
Q20.8.1 B	Q20.8.2 A	Q20.8.3 D	Q20.8.4 C	Q20.8.5 C	Q20.8.6 A
Q20.8.7 C	Q20.8.8 C	Q20.8.9 B	Q20.8.10 MTA		

YEAR

2019

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CY, ME-1, MN	10
Session 2	ME-2, XE, XL, TF	10
Session 3	CS, CH	10
Session 4	AE, AG, AR, BT, EY, GG, IN, MA, MT, PE, PH, PI, ST	10
Session 5	EC	10
Session 6	EE	10
Session 7	CE-1	10
Session 8	CE-2	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 02/02/19

Q19.1.1 MCQ 2019 1M

John Thomas, an _____ writer, passed away in 2018.

A. imminent **B.** prominent **C.** eminent **D.** dominant

Q19.1.2 MCQ 2019 1M

_____ I permitted him to leave, I wouldn't have had any problem with him being absent, _____ I?

A. Had, wouldn't **B.** Have, would

C. Had, would **D.** Have, wouldn't

Q19.1.3 MCQ 2019 1M

A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory?

A. 3.75 hours **B.** 4 hours and 15 mins

C. 8.5 hours **D.** 7.5 hours

Q19.1.4 MCQ 2019 1M

The sum and product of two integers are 26 and 165 respectively. The difference between these two integers is

A. 2 **B.** 3 **C.** 4 **D.** 6

Q19.1.5 MCQ 2019 1M

The minister avoided any mention of the issue of women's reservation in the private sector. He was accused of _____ the issue.

A. collaring **B.** skirting **C.** tying **D.** belting

Q19.1.6 MCQ 2019 2M

Under a certain legal system, prisoners are allowed to make one statement. If their statement turns out to be true then they are hanged. If the statement turns out to be false then they are shot. One prisoner made a statement and the judge had no option but to set him free. Which one of the following could be that statement?

A. I did not commit the crime

B. I committed the crime

C. I will be shot

D. You committed the crime

Q19.1.7 MCQ 2019 2M

A person divided an amount of Rs. 100,000 into two parts and invested in two different schemes. In one he got 10% profit and in the other he got 12%. If the profit percentages are interchanged with these investments he would have got Rs. 120 less. Find the ratio between his investments in the two schemes.

A. 9 : 16 **B.** 11 : 14 **C.** 37 : 63 **D.** 47 : 53

Q19.1.8 MCQ 2019 2M

Congo was named by Europeans. Congo's dictator Mobuto later changed the name of the country and the river to Zaire with the objective of Africanising names of persons and spaces. However, the name Zaire was a Portuguese alteration of *Nzadi o Nzere*, a local African term meaning 'River that swallows Rivers'. Zaire was the Portuguese name for the Congo river in the 16th and 17th centuries. Which one of the following statements can be inferred from the paragraph above?

A. Mobuto was not entirely successful in Africanising the name of his country

B. The term *Nzadi o Nzere* was of Portuguese origin

C. Mobuto's desire to Africanise names was prevented by the Portuguese

D. As a dictator Mobuto ordered the Portuguese to alter the name of the river to Zaire

Q19.1.9 MCQ 2019 2M

A firm hires employees at five different skill levels P, Q, R, S, T. The shares of employment at these skill levels of total employment in 2010 is given in the pie chart as shown. There were a total of 600 employees in 2010 and the total employment increased by 15% from 2010 to 2016. The total employment at skill levels P, Q and R remained unchanged during this period. If the employment at skill level S increased by 40% from 2010 to 2016, how many employees were there at skill level T in 2016?

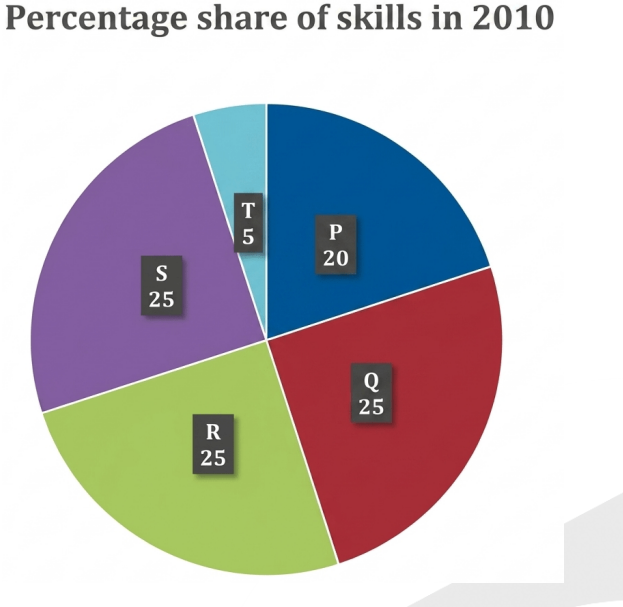


Fig. Q19.1.9

A. 30 **B.** 35 **C.** 60 **D.** 72

Q19.1.10 MCQ | 2019 | 2M

M and N had four children P, Q, R and S. Of them, only P and R were married. They had children X and Y respectively. If Y is a legitimate child of W, which one of the following statements is necessarily FALSE?

A. M is the grandmother of Y **B.** R is the father of Y

C. W is the wife of R **D.** W is the wife of P

Session : 2 | Afternoon | 02/02/19

Q19.2.1 MCQ | 2019 | 1M

Once the team of analysts identify the problem, we _____ in a better position to comment on the issue.
Which one of the following choices CANNOT fill the given blank?

A. will be **B.** were to be

C. are going to be **D.** might be

Q19.2.2 MCQ | 2019 | 1M

A final examination is the _____ of a series of evaluations that a student has to go through.

A. culmination **B.** consultation

C. desperation **D.** insinuation

Q19.2.3 MCQ | 2019 | 1M

If IMHO = JNIP; IDK = JEL; and SO = TP, then IDC = _____.

A. JDE **B.** JED **C.** JDC **D.** JCD

Q19.2.4 MCQ | 2019 | 1M

The product of three integers X, Y and Z is 192. Z is equal to 4 and P is equal to the average of X and Y. What is the minimum possible value of P?

A. 6 **B.** 7 **C.** 8 **D.** 9.5

Q19.2.5 MCQ | 2019 | 1M

Are there enough seats here? There are _____ people here than I expected.

A. many **B.** most **C.** least **D.** more

Q19.2.6 MCQ | 2019 | 2M

Fiscal deficit was 4% of the GDP in 2015 and that increased to 5% in 2016. If the GDP increased by 10% from 2015 to 2016, the percentage increase in the actual fiscal deficit is _____.

A. 37.50 **B.** 35.70 **C.** 25.00 **D.** 10.00

Q19.2.7 MCQ | 2019 | 2M

Two pipes P and Q can fill a tank in 6 hours and 9 hours respectively, while a third pipe R can empty the tank in 12 hours. Initially, P and R are open for 4 hours. Then P is closed and Q is opened. After 6 more hours R is closed. The total time taken to fill the tank (in hours) is _____.

A. 13.50 **B.** 14.50 **C.** 15.50 **D.** 16.50

Q19.2.8 MCQ | 2019 | 2M

While teaching a creative writing class in India, I was surprised at receiving stories from the students that were all set in distant places: in the American West with cowboys and in Manhattan penthouses with clinking ice cubes. This was, till an eminent Caribbean writer gave the writers in the once-colonised countries the confidence to see the shabby lives around them as worthy of being "told".
The writer of this passage is surprised by the creative writing assignments of his students, because _____.

A. Some of the students had written stories set in foreign places

B. None of the students had written stories set in India

C. None of the students had written about ice cubes and cowboys

D. Some of the students had written about ice cubes and cowboys

Q19.2.9 MCQ 2019 2M

Mola is a digital platform for taxis in a city. It offers three types of rides – Pool, Mini and Prime. The table below presents the number of rides for the past four months. The platform earns one US dollar per ride. What is the percentage share of revenue contributed by Prime to the total revenues of Mola, for the entire duration?

Type	Month			
	January	February	March	April
Pool	170	320	215	190
Mini	110	220	180	70
Prime	75	180	120	90

A. 16.24 **B.** 23.97 **C.** 25.86 **D.** 38.74

Q19.2.10 MCQ 2019 2M

X is an online media provider. By offering unlimited and exclusive online content at attractive prices for a loyalty membership, X is almost forcing its customers towards its loyalty membership. If its loyalty membership continues to grow at its current rate, within the next eight years more households will be watching X than cable television. Which one of the following statements can be inferred from the above paragraph?

A. Most households that subscribe to X's loyalty membership discontinue watching cable television

B. Non-members prefer to watch cable television

C. Cable television operators don't subscribe to X's loyalty membership

D. The X is cancelling accounts of non-members

Session : 3 | Forenoon | 03/02/19

Q19.3.1 MCQ 2019 1M

The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

A. break down **B.** break **C.** breaks down **D.** breaks

Q19.3.2 MCQ 2019 1M

The search engine's business model _____ around the fulcrum of trust.

A. revolves **B.** plays **C.** sinks **D.** bursts

Q19.3.3 MCQ 2019 1M

Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is _____.

A. 1 **B.** 2 **C.** 3 **D.** 6

Q19.3.4 MCQ 2019 1M

Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs 150 more. The cost of the gift was Rs. _____.

A. 666 **B.** 3000 **C.** 6000 **D.** 12000

Q19.3.5 MCQ 2019 1M

A court is to a judge as _____ is to a teacher.

A. a student **B.** a punishment

C. a syllabus **D.** a school

Q19.3.6 MCQ 2019 2M

The police arrested four criminals – P, Q, R and S. The criminals knew each other. They made the following statements:

P says "Q committed the crime."
Q says "S committed the crime."
R says "I did not do it."
S says "What Q said about me is false."

Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?

A. P **B.** R **C.** S **D.** Q

Q19.3.7 MCQ 2019 2M

In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _____.

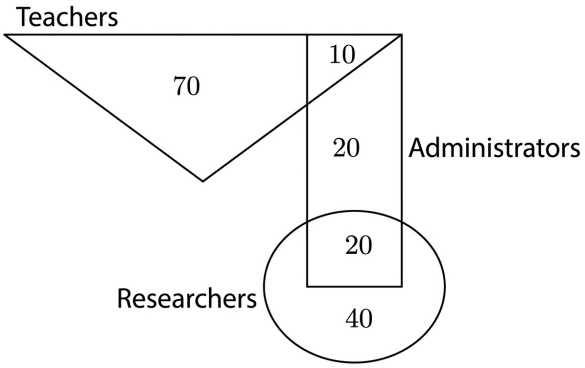


Fig. Q19.3.7

A. 0 to 15 **B.** 16 to 30 **C.** 31 to 45 **D.** 46 to 60

Q19.3.8 MCQ 2019 2M

"A recent High Court judgement has sought to dispel the idea of begging as a disease — which leads to its stigmatization and criminalization — and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net."
Which one of the following statements can be inferred from the given passage?

A. Beggars are lazy people who beg because they are unwilling to work

B. Beggars are created because of the lack of social welfare schemes

C. Begging is an offence that has to be dealt with firmly

D. Begging has to be banned because it adversely affects the welfare of the state

Q19.3.9 MCQ 2019 2M

In a college, there are three student clubs. Sixty students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in the Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all the clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is

A. 1000 **B.** 975 **C.** 900 **D.** 225

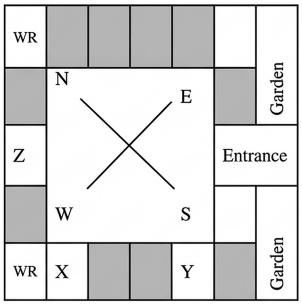
Q19.3.10 MCQ 2019 2M

Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.

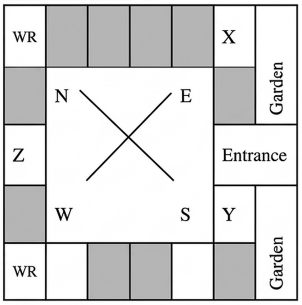
Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.
Request by Y: I want to live as far from the washrooms as possible, since I am very sensitive to smell.
Request by Z: I believe in Vaastu and so want to stay in the South-west wing.

The shaded rooms are already occupied. WR is washroom.

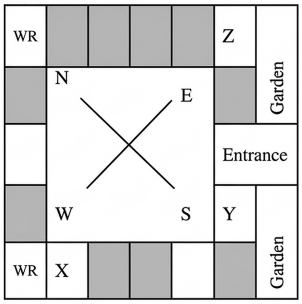
A.



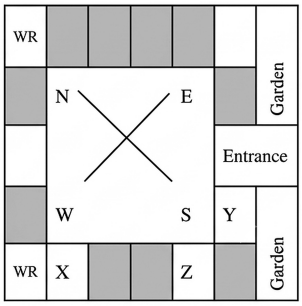
B.



C.



D.



Session : 4 | Afternoon | 03/02/19

Q19.4.1 MCQ 2019 1M

The fishermen, _____ the flood victims owed their lives, were rewarded by the government.

A. whom **B.** to which **C.** to whom **D.** that

Q19.4.2 MCQ 2019 1M

Some students were not involved in the strike.
If the above statement is true, which of the following conclusions is/are logically necessary?

1. Some who were involved in the strike were students.
2. No student was involved in the strike.
3. At least one student was involved in the strike.
4. Some who were not involved in the strike were students.

A. 1 and 2 **B.** 3 **C.** 4 **D.** 2 and 3

Q19.4.3 MCQ 2019 1M

The radius as well as the height of a circular cone increases by 10%. The percentage increase in its volume is_____.

A. 17.1 **B.** 21.0 **C.** 33.1 **D.** 72.8

Q19.4.4 MCQ 2019 1M

Five numbers 10, 7, 5, 4 and 2 are to be arranged in a sequence from left to right following the directions given below:

1. No two odd or even numbers are next to each other.
2. The second number from the left is exactly half of the left-most number.
3. The middle number is exactly twice the right-most number.

Which is the second number from the right?

A. 2 **B.** 4 **C.** 7 **D.** 10

Q19.4.5 MCQ 2019 1M

Until Iran came along, India had never been _____ in kabaddi.

A. defeated **B.** defeating **C.** defeat **D.** defeatist

Q19.4.6 MCQ 2019 2M

Since the last one year, after a 125 basis point reduction in repo rate by the Reserve Bank of India, banking institutions have been making a demand to reduce interest rates on small saving schemes. Finally, the government announced yesterday a reduction in interest rates on small saving schemes to bring them on par with fixed deposit interest rates.
Which one of the following statements can be inferred from the given passage?

A. Whenever the Reserve Bank of India reduces the repo rate, the interest rates on small saving schemes are also reduced

B. Interest rates on small saving schemes are always maintained on par with fixed deposit interest rates

C. The government sometimes takes into consideration the demands of banking institutions before reducing the interest rates on small saving schemes

D. A reduction in interest rates on small saving schemes follow only after a reduction in repo rate by the Reserve Bank of India

Q19.4.7 MCQ 2019 2M

In a country of 1400 million population, 70% own mobile phones. Among the mobile phone owners, only 294 million access the Internet. Among these Internet users, only half buy goods from e-commerce portals. What is the percentage of these buyers in the country?

A. 10.50 **B.** 14.70 **C.** 15.00 **D.** 50.00

Q19.4.8 MCQ 2019 2M

The nomenclature of Hindustani music has changed over the centuries. Since the medieval period *dhrupad* styles were identified as *baanis*. Terms like *gayaki* and *baaj* were used to refer to vocal and instrumental styles, respectively. With the institutionalization of music education the term *gharana* became acceptable. *Gharana* originally referred to hereditary musicians from a particular lineage, including disciples and grand disciples.
Which one of the following pairings is NOT correct?

A. *dhrupad*, *baani* **B.** *gayaki*, vocal

C. *baaj*, institution **D.** *gharana*, lineage

Q19.4.9 MCQ 2019 2M

Two trains started at 7AM from the same point. The first train travelled north at a speed of 80 km/h and the second train travelled south at a speed of 100 km/h. The time at which they were 540 km apart is ____ AM.

A. 9 **B.** 10 **C.** 11 **D.** 11.30

Q19.4.10 MCQ 2019 2M

"I read somewhere that in ancient times the prestige of a kingdom depended upon the number of taxes that it was able to levy on its people. It was very much like the prestige of a head-hunter in his own community."
Based on the paragraph above, the prestige of a head-hunter depended upon _____

A. the prestige of the kingdom

B. the prestige of the heads

C. the number of taxes he could levy

D. the number of heads he could gather

Session : 5 | Forenoon | 09/02/19

Q19.5.1 MCQ 2019 1M

The strategies that the company _____ to sell its products _____ house-to-house marketing.

A. use, includes **B.** uses, include

C. used, includes **D.** uses, including

Q19.5.2 MCQ 2019 1M

The boat arrived _____ dawn.

A. in **B.** at **C.** on **D.** under

Q19.5.3 MCQ 2019 1M

It would take one machine 4 hours to complete a production order and another machine 2 hours to complete the same order. If both machines work simultaneously at their respective constant rates, the time taken to complete the same order is _____ hours.

A. 2/3 **B.** 3/4 **C.** 4/3 **D.** 7/3

Q19.5.4 MCQ 2019 1M

Five different books (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is _____.

A. 2 **B.** 6 **C.** 12 **D.** 120

Q19.5.5 MCQ 2019 1M

When he did not come home, she _____ him lying dead on the roadside somewhere.

A. concluded **B.** looked **C.** notice **D.** pictured

Q19.5.6 MCQ 2019 2M

Four people are standing in a line facing you. They are Rahul, Mathew, Seema and Lohit. One is an engineer, one is a doctor, one a teacher and another a dancer. You are told that:

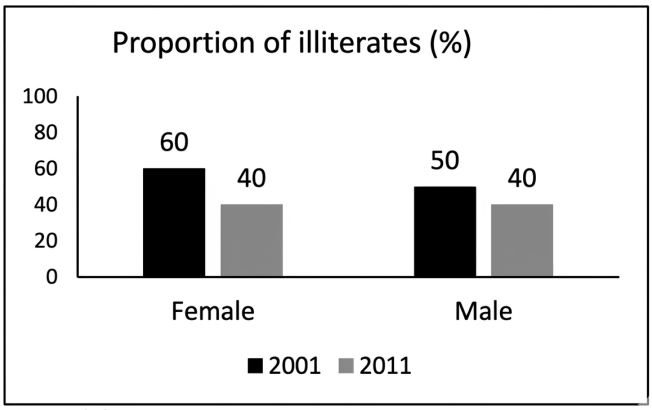
1. Mathew is not standing next to Seema
2. There are two people standing between Lohit and the engineer
3. Rahul is not a doctor
4. The teacher and the dancer are standing next to each other
5. Seema is turning to her right to speak to the doctor standing next to her

Who among them is an engineer?

A. Seema **B.** Lohit **C.** Rahul **D.** Mathew

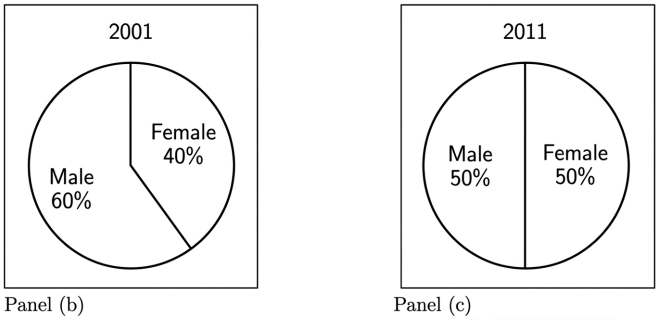
Q19.5.7 MCQ 2019 2M

The bar graph in Panel (a) shows the proportion of male and female illiterates in 2001 and 2011. The proportions of males and females in 2001 and 2011 are given in Panel (b) and (c), respectively. The total population did not change during this period.
The percentage increase in the total number of literates from 2001 to 2011 is _____.



Panel (a)

Fig. Q19.5.7



Panel (b) Panel (c)

Fig. Q19.5.7

A. 30.43 **B.** 33.43 **C.** 34.43 **D.** 35.43

Q19.5.8 MCQ 2019 2M

"Indian history was written by British historians – extremely well documented and researched, but not always impartial. History had to serve its purpose: Everything was made subservient to the glory of the Union Jack. Latter-day Indian scholars presented a contrary picture."
From the text above, we can infer that:
Indian history written by British historians _____

A. was well documented and not researched but was always biased

B. was not well documented and researched and was always biased

C. was well documented and researched but was sometimes biased

D. was not well documented and researched and was sometimes biased

Q19.5.9 MCQ 2019 2M

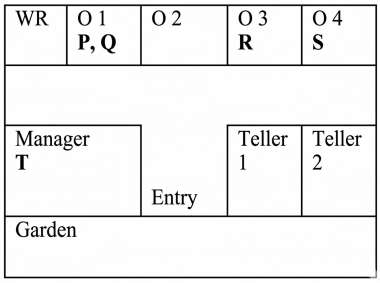
Two design consultants, P and Q, started working from 8 AM for a client. The client budgeted a total of USD 3000 for the consultants. P stopped working when the hour hand moved by 210 degrees on the clock. Q stopped working when the hour hand moved by 240 degrees. P took two tea breaks of 15 minutes each during her shift, but took no lunch break. Q took only one lunch break for 20 minutes, but no tea breaks. The market rate for consultants is USD 200 per hour and breaks are not paid. After paying the consultants, the client shall have USD _____ remaining in the budget.

A. 000.00 **B.** 166.67 **C.** 300.00 **D.** 433.33

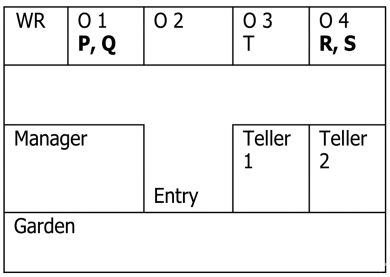
Q19.5.10 MCQ 2019 2M

Five people P, Q, R, S and T work in a bank. P and Q don't like each other but have to share an office till T gets a promotion and moves to the big office next to the garden. R, who is currently sharing an office with T wants to move to the adjacent office with S, the handsome new intern. Given the floor plan, what is the current location of Q, R and T? (O = Office, WR = Washroom)

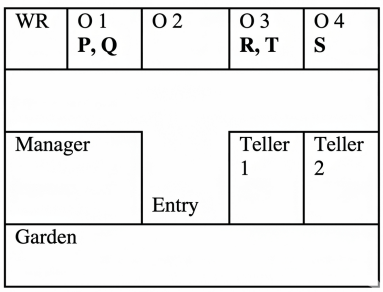
A.



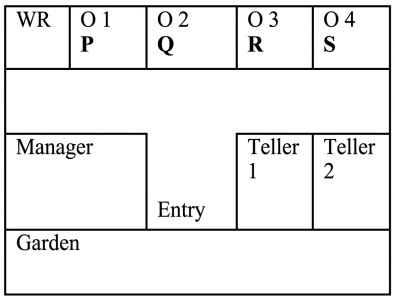
B.



C.



D.



Session : 6 | Afternoon | 09/02/19

Q19.6.1 MCQ 2019 1M

I am not sure if the bus that has been booked will be able to _____ all the students.

A. sit **B.** deteriorate **C.** fill **D.** accommodate

Q19.6.2 MCQ 2019 1M

The passengers were angry _____ the airline staff about the delay.

A. on **B.** about **C.** with **D.** towards

Q19.6.3 MCQ 2019 1M

The missing number in the given sequence 343, 1331, ____, 4913 is

A. 3375 **B.** 2744 **C.** 2197 **D.** 4096

Q19.6.4 MCQ 2019 1M

It takes two hours for a person X to mow the lawn. Y can mow the same lawn in four hours. How long (in minutes) will it take X and Y, if they work together to mow the lawn?

A. 60 **B.** 80 **C.** 90 **D.** 120

Q19.6.5 MCQ 2019 1M

Newspapers are a constant source of delight and recreation for me. The ____ trouble is that I read ____ many of them.

A. even, quite **B.** even, too **C.** only, quite **D.** only, too

Q19.6.6 MCQ 2019 2M

How many integers are there between 100 and 1000 all of whose digits are even?

A. 60 **B.** 80 **C.** 100 **D.** 90

Q19.6.7 MCQ 2019 2M

The ratio of the number of boys and girls who participated in an examination is 4:3. The total percentage of candidates who passed the examination is 80 and the percentage of girls who passed is 90. The percentage of boys who passed is _____.

A. 55.50 **B.** 72.50 **C.** 80.50 **D.** 90.00

Q19.6.8 MCQ 2019 2M

An award-winning study by a group of researchers suggests that men are as prone to buying on impulse as women but women feel more guilty about shopping.
Which one of the following statements can be inferred from the given text?

A. Some men and women indulge in buying on impulse

B. All men and women indulge in buying on impulse

C. Few men and women indulge in buying on impulse

D. Many men and women indulge in buying on impulse

Q19.6.9 MCQ 2019 2M

Given two sets $X = \{1, 2, 3\}$ and $Y = \{2, 3, 4\}$, we construct a set Z of all possible fractions where the numerators belong to set X and the denominators belong to set Y. The product of elements having minimum and maximum values in the set Z is _____.

A. 1/12 **B.** 1/8 **C.** 1/6 **D.** 3/8

Q19.6.10 MCQ 2019 2M

Consider five people – Mita, Ganga, Rekha, Lakshmi and Sana. Ganga is taller than both Rekha and Lakshmi. Lakshmi is taller than Sana. Mita is taller than Ganga.
Which of the following conclusions are true?

1. Lakshmi is taller than Rekha
2. Rekha is shorter than Mita
3. Rekha is taller than Sana
4. Sana is shorter than Ganga

A. 1 and 3 **B.** 3 only **C.** 2 and 4 **D.** 1 only

Session : 7 | Forenoon | 10/02/19

Q19.7.1 MCQ 2019 1M

The lecture was attended by quite _____ students, so the hall was not very _____.

A. a few, quite **B.** few, quiet

C. a few, quiet **D.** few, quite

Q19.7.2 MCQ 2019 1M

They have come a long way in _____ trust among the users.

A. creating **B.** created **C.** creation **D.** create

Q19.7.3 MCQ 2019 1M

On a horizontal ground, the base of a straight ladder is 6 m away from the base of a vertical pole. The ladder makes an angle of 45° to the horizontal. If the ladder is resting at a point located at one-fifth of the height of the pole from the bottom, the height of the pole is ____ meters.

A. 15 **B.** 25 **C.** 30 **D.** 35

Q19.7.4 MCQ 2019 1M

If E = 10; J = 20; O = 30; and T = 40, what will be P + E + S + T?

A. 51 **B.** 82 **C.** 120 **D.** 164

Q19.7.5 MCQ 2019 1M

The CEO's decision to quit was as shocking to the Board as it was to _____.

A. I **B.** me **C.** my **D.** myself

Q19.7.6 MCQ 2019 2M

The new cotton technology, Bollgard-II, with herbicide tolerant traits has developed into a thriving business in India. However, the commercial use of this technology is not legal in India. Notwithstanding that, reports indicate that the herbicide tolerant Bt cotton had been purchased by farmers at an average of Rs 200 more than the control price of ordinary cotton, and planted in 15% of the cotton growing area in the 2017 Kharif season.
Which one of the following statements can be inferred from the given passage?

A. Farmers want to access the new technology if India benefits from it

B. Farmers want to access the new technology even if it is not legal

C. Farmers want to access the new technology for experimental purposes

D. Farmers want to access the new technology by paying high price

Q19.7.7 MCQ 2019 2M

In a sports academy of 300 people, 105 play only cricket, 70 play only hockey, 50 play only football, 25 play both cricket and hockey, 15 play both hockey and football and 30 play both cricket and football. The rest of them play all three sports. What is the percentage of people who play at least two sports?

A. 23.30 **B.** 25.00 **C.** 28.00 **D.** 50.00

Q19.7.8 MCQ 2019 2M

"The increasing interest in tribal characters might be a mere coincidence, but the timing is of interest. None of this, though, is to say that the tribal hero has arrived in Hindi cinema, or that the new crop of characters represents the acceptance of the tribal character in the industry. The films and characters are too few to be described as a pattern."
What does the word 'arrived' mean in the paragraph above?

A. reached a terminus **B.** came to a conclusion

C. attained a status **D.** went to a place

Q19.7.9 MCQ 2019 2M

A square has sides 5 cm smaller than the sides of a second square. The area of the larger square is four times the area of the smaller square. The side of the larger square is ___ cm.

A. 18.50 **B.** 15.10 **C.** 10.00 **D.** 8.50

Q19.7.10 MCQ 2019 2M

P, Q, R, S and T are related and belong to the same family. P is the brother of S. Q is the wife of P. R and T are the children of the siblings P and S respectively. Which one of the following statements is necessarily FALSE?

A. S is the aunt of R **B.** S is the aunt of T

C. S is the sister-in-law of Q **D.** S is the brother of P

Session : 8 | Afternoon | 10/02/19

Q19.8.1 MCQ 2019 1M

Daytime temperatures in Delhi can _____ 40°C.

A. get **B.** stand **C.** reach **D.** peak

Q19.8.2 MCQ 2019 1M

The growth rate of ABC Motors in 2017 was the same _____ XYZ Motors in 2016.

A. as off **B.** as those of **C.** as that off **D.** as that of

Q19.8.3 MCQ 2019 1M

Suresh wanted to lay a new carpet in his new mansion with an area of 70×55 sq. mts. However an area of 550 sq. mts. had to be left out for flower pots. If the cost of carpet is Rs. 50 per sq. mts., how much money (in Rs.) will be spent by Suresh for the carpet now?

A. Rs. 1,65,000 **B.** Rs. 1,92,500

C. Rs. 2,75,000 **D.** Rs. 1,27,500

Q19.8.4 MCQ 2019 1M

A retaining wall with measurements $30\text{ m} \times 12\text{ m} \times 6\text{ m}$ was constructed with bricks of dimensions $8\text{ cm} \times 6\text{ cm} \times 6\text{ cm}$. If 60% of the wall consists of bricks, the number of bricks used for the construction is ____ lakhs.

A. 30 **B.** 40 **C.** 45 **D.** 75

Q19.8.5 MCQ 2019 1M

Hima Das was ____ only Indian athlete to win ____ gold for India.

A. the, many **B.** the, a **C.** an, a **D.** an, the

Q19.8.6 MCQ 2019 2M

Mohan, the manager, wants his four workers to work in pairs. No pair should work for more than 5 hours. Ram and John have worked together for 5 hours. Krishna and Amir have worked as a team for 2 hours. Krishna does not want to work with Ram. Whom should Mohan allot to work with John, if he wants all the workers to continue working?

A. Amir **B.** Krishna **C.** Ram **D.** None of the three

Q19.8.7 MCQ 2019 2M

Population of state X increased by x% and the population of state Y increased by y% from 2001 to 2011. Assume that x is greater than y. Let P be the ratio of the population of state X to state Y in a given year. The percentage increase in P from 2001 to 2011 is _____.

A. $\frac{x}{y}$ **B.** $x - y$ **C.** $\frac{100(x-y)}{100+x}$ **D.** $\frac{100(x-y)}{100+y}$

Q19.8.8 MCQ 2019 2M

The Newspaper reports that over 500 hectares of tribal land spread across 28 tribal settlements in Mohimitampuram forest division have already been "alienated". A top forest official said, "First the tribals are duped out of their land holdings. Second, the families thus rendered landless are often forced to encroach further into the forests".

On the basis of the information available in the paragraph, _____ is/are responsible for duping the tribals.

A. forest officials **B.** landless families

C. The Newspaper **D.** it cannot be inferred who

Q19.8.9 MCQ 2019 2M

An oil tank can be filled by pipe X in 5 hours and pipe Y in 4 hours, each pump working on its own. When the oil tank is full and the drainage hole is open, the oil is drained in 20 hours. If initially the tank was empty and someone started the two pumps together but left the drainage hole open, how many hours will it take for the tank to be filled? (Assume that the rate of drainage is independent of the Head)

A. 1.50 **B.** 2.00 **C.** 2.50 **D.** 4.00

Q19.8.10 MCQ 2019 2M

"Popular Hindi fiction, despite – or perhaps because of – its wide reach, often does not appear in our cinema. As ideals that viewers are meant to look up to rather than identify with, Hindi film protagonists usually read books of aspirational value: textbooks, English books, or high value literature."

Which one of the following CANNOT be inferred from the paragraph above?

A. Though popular Hindi fiction has wide reach, it often does not appear in the movies

B. Protagonists in Hindi movies, being ideals for viewers, read only books of aspirational value

C. Textbooks, English books or high literature have aspirational value, but not popular Hindi fiction

D. People do not look up to writers of textbooks, English books, or high value literature

PrepFusion

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q19.1.1 C	Q19.1.2 C	Q19.1.3 D	Q19.1.4 C	Q19.1.5 B	Q19.1.6 C
Q19.1.7 D	Q19.1.8 A	Q19.1.9 C	Q19.1.10 D		
Session : 2					
Q19.2.1 B	Q19.2.2 A	Q19.2.3 B	Q19.2.4 MTA	Q19.2.5 D	Q19.2.6 A
Q19.2.7 B	Q19.2.8 B	Q19.2.9 B	Q19.2.10 A		
Session : 3					
Q19.3.1 C	Q19.3.2 A	Q19.3.3 B	Q19.3.4 C	Q19.3.5 D	Q19.3.6 B
Q19.3.7 C	Q19.3.8 B	Q19.3.9 C	Q19.3.10 D		
Session : 4					
Q19.4.1 C	Q19.4.2 C	Q19.4.3 C	Q19.4.4 C	Q19.4.5 A	Q19.4.6 C
Q19.4.7 A	Q19.4.8 C	Q19.4.9 B	Q19.4.10 D		
Session : 5					
Q19.5.1 B	Q19.5.2 B	Q19.5.3 C	Q19.5.4 B	Q19.5.5 D	Q19.5.6 D
Q19.5.7 A	Q19.5.8 C	Q19.5.9 B	Q19.5.10 C		
Session : 6					
Q19.6.1 D	Q19.6.2 C	Q19.6.3 C	Q19.6.4 B	Q19.6.5 D	Q19.6.6 C
Q19.6.7 B	Q19.6.8 A	Q19.6.9 D	Q19.6.10 C		
Session : 7					
Q19.7.1 C	Q19.7.2 A	Q19.7.3 C	Q19.7.4 C	Q19.7.5 B	Q19.7.6 B
Q19.7.7 B	Q19.7.8 C	Q19.7.9 C	Q19.7.10 B		
Session : 8					
Q19.8.1 C	Q19.8.2 D	Q19.8.3 A	Q19.8.4 C	Q19.8.5 B	Q19.8.6 B
Q19.8.7 D	Q19.8.8 D	Q19.8.9 C	Q19.8.10 B or D		

YEAR

2018

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	ME-1, EY, PE, XE, XL	10
Session 2	ME-2, AE, MA ,PI	10
Session 3	CS, MN	10
Session 4	AG, AR, BT, CH, CY, GG, IN, MT, PH, TF	10
Session 5	EC	10
Session 6	EE	10
Session 7	CE-1	10
Session 8	CE-2	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 03/02/18

Q18.1.1 MCQ 2018 1M

"Going by the _____ that many hands make light work, the school _____ involved all the students in the task."
The words that best fill the blanks in the above sentence are

A. principle, principal **B.** principal, principle

C. principle, principle **D.** principal, principal

Q18.1.2 MCQ 2018 1M

"Her _____ should not be confused with miserliness; she is ever willing to assist those in need."
The word that best fills the blank in the above sentence is

A. cleanliness **B.** punctuality

C. frugality **D.** greatness

Q18.1.3 MCQ 2018 1M

Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machines to make 100 toys?

A. 1 **B.** 7 **C.** 100 **D.** 700

Q18.1.4 MCQ 2018 1M

A rectangle becomes a square when its length and breadth are reduced by 10 m and 5 m, respectively. During this process, the rectangle loses 650 m^2 of area. What is the area of the original rectangle in square meters?

A. 1125 **B.** 2250 **C.** 2924 **D.** 4500

Q18.1.5 MCQ 2018 1M

A number consists of two digits. The sum of the digits is 9. If 45 is subtracted from the number, its digits are interchanged. What is the number?

A. 63 **B.** 72 **C.** 81 **D.** 90

Q18.1.6 MCQ 2018 2M

For integers a, b and c, what would be the minimum and maximum values respectively of $a+b+c$ if $\log|a|+\log|b|+\log|c| = 0$?

A. -3 and 3 **B.** -1 and 1 **C.** -1 and 3 **D.** 1 and 3

Q18.1.7 MCQ 2018 2M

Given that a and b are integers and $a + a^2 b^3$ is odd, which one of the following statements is correct?

A. a and b are both odd **B.** a and b are both even

C. a is even and b is odd **D.** a is odd and b is even

Q18.1.8 MCQ 2018 2M

From the time the front of a train enters a platform, it takes 25 seconds for the back of the train to leave the platform, while travelling at a constant speed of 54 km/h. At the same speed, it takes 14 seconds to pass a man running at 9 km/h in the same direction as the train. What is the length of the train and that of the platform in meters, respectively?

A. 210 and 140 **B.** 162.5 and 187.5

C. 245 and 130 **D.** 175 and 200

Q18.1.9 MCQ 2018 2M

Which of the following functions describe the graph shown in the below figure?

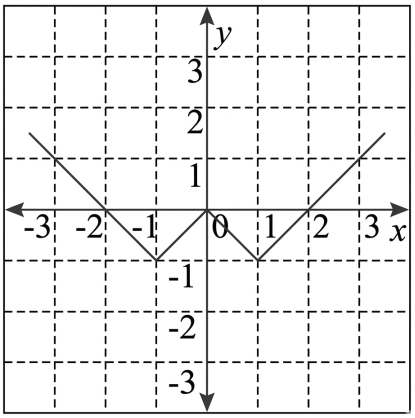


Fig. Q18.1.9

A. $y = \big||x| + 1\big| - 2$ **B.** $y = \big||x| - 1\big| - 1$

C. $y = \big||x| + 1\big| - 1$ **D.** $y = \big||x - 1| - 1\big|$

Q18.1.10 MCQ 2018 2M

Consider the following three statements:

(i) Some roses are red.
(ii) All red flowers fade quickly.
(iii) Some roses fade quickly.

Which of the following statements can be logically inferred from the above statements?

A. If (i) is true and (ii) is false, then (iii) is false.

B. If (i) is true and (ii) is false, then (iii) is true.

C. If (i) and (ii) are true, then (iii) is true.

D. If (i) and (ii) are false, then (iii) is false.

Session : 2 | Afternoon | 03/02/18

Q18.2.1 MCQ 2018 1M

"The dress _____ her so well that they all immediately _____ her on her appearance."
The words that best fill the blanks in the above sentence are

A. complemented, complemented

B. complimented, complemented

C. complimented, complimented

D. complemented, complimented

Q18.2.2 MCQ 2018 1M

"The judge's standing in the legal community, though shaken by false allegations of wrongdoing, remained ____."
The word that best fills the blank in the above sentence is

A. undiminished **B.** damaged

C. illegal **D.** uncertain

Q18.2.3 MCQ 2018 1M

Find the missing group of letters in the following series:
BC, FGH, LMNO, _____

A. UVWXY **B.** TUVWX **C.** STUVW **D.** RSTUV

Q18.2.4 MCQ 2018 1M

The perimeters of a circle, a square and an equilateral triangle are equal.
Which one of the following statements is true?

A. The circle has the largest area.

B. The square has the largest area.

C. The equilateral triangle has the largest area.

D. All the three shapes have the same area.

Q18.2.5 MCQ 2018 1M

The value of the expression $\frac{1}{1+\log_u vw} + \frac{1}{1+\log_v wu} + \frac{1}{1+\log_w uv}$ is _____.

A. -1 **B.** 0 **C.** 1 **D.** 3

Q18.2.6 MCQ 2018 2M

Forty students watched films A, B and C over a week. Each student watched either only one film or all three. Thirteen students watched film A, sixteen students watched film B and nineteen students watched film C.
How many students watched all three films?

A. 0 **B.** 2 **C.** 4 **D.** 8

Q18.2.7 MCQ 2018 2M

A wire would enclose an area of 1936 m^2, if it is bent into a square. The wire is cut into two pieces. The longer piece is thrice as long as the shorter piece. The long and the short pieces are bent into a square and a circle, respectively. Which of the following choices is closest to the sum of the areas enclosed by the two pieces in square meters?

A. 1096 **B.** 1111 **C.** 1243 **D.** 2486

Q18.2.8 MCQ 2018 2M

A contract is to be completed in 52 days and 125 identical robots were employed, each operational for 7 hours a day. After 39 days, five-seventh of the work was completed. How many additional robots would be required to complete the work on time, if each robot is now operational for 8 hours a day?

A. 50 **B.** 89 **C.** 146 **D.** 175

Q18.2.9 MCQ 2018 2M

A house has a number which needs to be identified. The following three statements are given that can help in identifying the house number.

(i) If the house number is a multiple of 3, then it is a number from 50 to 59.
(ii) If the house number is NOT a multiple of 4, then it is a number from 60 to 69.
(iii) If the house number is NOT a multiple of 6, then it is a number from 70 to 79.

What is the house number?

A. 54 **B.** 65 **C.** 66 **D.** 76

Q18.2.10 MCQ 2018 2M

An unbiased coin is tossed six times in a row and four different such trials are conducted. One trial implies six tosses of the coin. If H stands for head and T stands for tail, the following are the observations from the four trials:
(1) HTHTTH (2) TTHHHT (3) HTTHHT
(4) HHHHT ___ ___
Which statement describing the last two coin tosses of the fourth trial has the highest probability of being correct?

A. Two T will occur.

B. One H and one T will occur.

C. Two H will occur.

D. One H will be followed by one T.

Session : 3 | Forenoon | 04/02/18

Q18.3.1 MCQ 2018 1M

"From where are they bringing their books? _____ bringing _____ books from _____."
The words that best fill the blanks in the above sentence are

A. Their, they're, there **B.** They're, their, there

C. There, their, they're **D.** They're, there, there

Q18.3.2 MCQ 2018 1M

"A _____ investigation can sometimes yield new facts, but typically organized ones are more successful."
The word that best fills the blank in the above sentence is

A. meandering **B.** timely

C. consistent **D.** systematic

Q18.3.3 MCQ 2018 1M

The area of a square is d. What is the area of the circle which has the diagonal of the square as its diameter?

A. πd **B.** πd^2 **C.** $\frac{1}{4}\pi d^2$ **D.** $\frac{1}{2}\pi d$

Q18.3.4 MCQ 2018 1M

What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

A. 3047 **B.** 6047 **C.** 7987 **D.** 63847

Q18.3.5 MCQ 2018 1M

What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, ____, 0

A. 2880 **B.** 1440 **C.** 720 **D.** 0

Q18.3.6 MCQ 2018 2M

In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs 750 to each male senior citizen in the town and Rs 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $8/9^{th}$ of the eligible men and $2/3^{rd}$ of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

A. 1,50,000 **B.** 2,00,000 **C.** 1,75,000 **D.** 1,51,000

Q18.3.7 MCQ 2018 2M

If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}$, $q^{-y} = \frac{1}{r}$, $r^{-z} = \frac{1}{p}$, what is the value of the product xyz?

A. -1 **B.** $\frac{1}{pqr}$ **C.** 1 **D.** pqr

Q18.3.8 MCQ 2018 2M

In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

A. 2:3 **B.** 1:1 **C.** 3:2 **D.** 2:1

Q18.3.9 MCQ 2018 2M

In the figure below, $\angle DEC + \angle BFC$ is equal to _____.

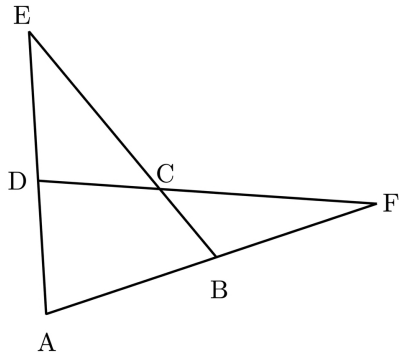


Fig. Q18.3.9

A. $\angle BCD - \angle BAD$ **B.** $\angle BAD + \angle BCF$

C. $\angle BAD + \angle BCD$ **D.** $\angle CBA + \angle ADC$

Q18.3.10 MCQ 2018 2M

A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

A. Three green faces and four red faces.

B. Four green faces and three red faces.

C. Five green faces and two red faces.

D. Six green faces and one red face.

Session : 4 | Afternoon | 04/02/18

Q18.4.1 MCQ 2018 1M

"When she fell down the _____, she received many _____ but little help."
The words that best fill the blanks in the above sentence are

A. stairs, stares **B.** stairs, stairs

C. stares, stairs **D.** stares, stares

Q18.4.2 MCQ 2018 1M

"In spite of being warned repeatedly, he failed to correct his _____ behaviour."
The word that best fills the blank in the above sentence is

A. rational **B.** reasonable **C.** errant **D.** good

Q18.4.3 MCQ 2018 1M

For $0 \le x \le 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _____.

A. $\left(0, \frac{\pi}{2}\right)$ **B.** $\left(\frac{\pi}{2}, \pi\right)$ **C.** $\left(\pi, \frac{3\pi}{2}\right)$ **D.** $\left(\frac{3\pi}{2}, 2\pi\right)$

Q18.4.4 MCQ 2018 1M

The area of an equilateral triangle is $\sqrt{3}$. What is the perimeter of the triangle?

A. 2 **B.** 4 **C.** 6 **D.** 8

Q18.4.5 MCQ 2018 1M

Arrange the following three-dimensional objects in the descending order of their volumes:

(i) A cuboid with dimensions 10 cm, 8 cm and 6 cm
(ii) A cube of side 8 cm
(iii) A cylinder with base radius 7 cm and height 7 cm
(iv) A sphere of radius 7 cm

A. (i), (ii), (iii), (iv) **B.** (ii), (i), (iv), (iii)

C. (iii), (ii), (i), (iv) **D.** (iv), (iii), (ii), (i)

Q18.4.6 MCQ 2018 2M

An automobile travels from city A to city B and returns to city A by the same route. The speed of the vehicle during the onward and return journeys were constant at 60 km/h and 90 km/h, respectively. What is the average speed in km/h for the entire journey?

A. 72 **B.** 73 **C.** 74 **D.** 75

Q18.4.7 MCQ 2018 2M

A set of 4 parallel lines intersect with another set of 5 parallel lines. How many parallelograms are formed?

A. 20 **B.** 48 **C.** 60 **D.** 72

Q18.4.8 MCQ 2018 2M

To pass a test, a candidate needs to answer at least 2 out of 3 questions correctly. A total of 6,30,000 candidates appeared for the test. Question A was correctly answered by 3,30,000 candidates. Question B was answered correctly by 2,50,000 candidates. Question C was answered correctly by 2,60,000 candidates. Both questions A and B were answered correctly by 1,00,000 candidates. Both questions B and C were answered correctly by 90,000 candidates. Both questions A and C were answered correctly by 80,000 candidates. If the number of students answering all questions correctly is the same as the number answering none, how many candidates failed to clear the test?

A. 30,000 **B.** 2,70,000 **C.** 3,90,000 **D.** 4,20,000

Q18.4.9 MCQ 2018 2M

If $x^2 + x - 1 = 0$ what is the value of $x^4 + \frac{1}{x^4}$?

A. 1 **B.** 5 **C.** 7 **D.** 9

Q18.4.10 MCQ 2018 2M

In a detailed study of annual crow births in India, it was found that there was relatively no growth during the period 2002 to 2004 and a sudden spike from 2004 to 2005. In another unrelated study, it was found that the revenue from cracker sales in India which remained fairly flat from 2002 to 2004, saw a sudden spike in 2005 before declining again in 2006. The solid line in the graph below refers to annual sale of crackers and the dashed line refers to the annual crow

births in India. Choose the most appropriate inference from the above data.

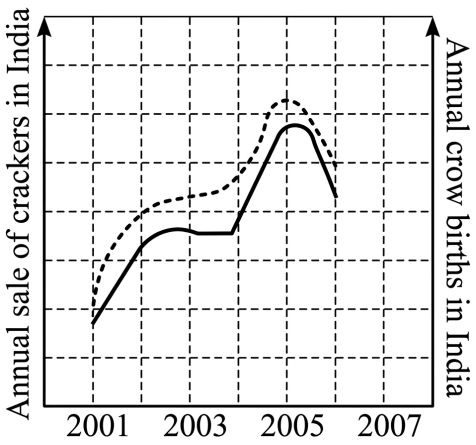


Fig. Q18.4.10

A. There is a strong correlation between crow birth and cracker sales.

B. Cracker usage increases crow birth rate.

C. If cracker sale declines, crow birth will decline.

D. Increased birth rate of crows will cause an increase in the sale of crackers.

Session : 5 | Forenoon | 10/02/18

Q18.5.1 MCQ 2018 1M

"By giving him the last ____ of the cake, you will ensure lasting ____ in our house today."
The words that best fill the blanks in the above sentence are

A. peas, piece **B.** piece, peace

C. peace, piece **D.** peace, peas

Q18.5.2 MCQ 2018 1M

"Even though there is a vast scope for its _____, tourism has remained a/an _____ area."
The words that best fill the blanks in the above sentence are

A. improvement, neglected **B.** rejection, approved

C. fame, glum **D.** interest, disinterested

Q18.5.3 MCQ 2018 1M

If the number 715■423 is divisible by 3 (■ denotes the missing digit in the thousandths place), then the smallest whole number in the place of ■ is ____.

A. 0 **B.** 2 **C.** 5 **D.** 6

Q18.5.4 MCQ 2018 1M

What is the value of $1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \cdots$?

A. 2 **B.** $\frac{7}{4}$ **C.** $\frac{3}{2}$ **D.** $\frac{4}{3}$

Q18.5.5 MCQ 2018 1M

A 1.5 m tall person is standing at a distance of 3 m from a lamp post. The light from the lamp at the top of the post casts her shadow. The length of the shadow is twice her height. What is the height of the lamp post in meters?

A. 1.5 **B.** 3 **C.** 4.5 **D.** 6

Q18.5.6 MCQ 2018 2M

Leila aspires to buy a car worth Rs. 10,00,000 after 5 years. What is the minimum amount in Rupees that she should deposit now in a bank which offers 10% annual rate of interest, if the interest was compounded annually?

A. 5,00,000 **B.** 6,21,000 **C.** 6,66,667 **D.** 7,50,000

Q18.5.7 MCQ 2018 2M

Two alloys A and B contain gold and copper in the ratios of 2:3 and 3:7 by mass, respectively. Equal masses of alloys A and B are melted to make an alloy C. The ratio of gold to copper in alloy C is ____.

A. 5:10 **B.** 7:13 **C.** 6:11 **D.** 9:13

Q18.5.8 MCQ 2018 2M

The Cricket Board has long recognized John's potential as a leader of the team. However, his on-field temper has always been a matter of concern for them since his junior days. While this aggression has filled stadia with die-hard fans, it has taken a toll on his own batting. Until recently, it appeared that he found it difficult to convert his aggression into big scores. Over the past three seasons though, that picture of John has been replaced by a cerebral, calculative and successful batsman-captain. After many years, it appears that the team has finally found a complete captain.
Which of the following statements can be logically inferred from the above paragraph?

(i) Even as a junior cricketer, John was considered a good captain.
(ii) Finding a complete captain is a challenge.
(iii) Fans and the Cricket Board have differing views on what they want in a captain.
(iv) Over the past three seasons John has accumulated big scores.

A. (i), (ii) and (iii) only **B.** (iii) and (iv) only

C. (ii) and (iv) only **D.** (i), (ii), (iii) and (iv)

Q18.5.9 MCQ 2018 2M

A cab was involved in a hit and run accident at night. You are given the following data about the cabs in the city and the accident.

(i) 85% of cabs in the city are green and the remaining cabs are blue.
(ii) A witness identified the cab involved in the accident as blue.
(iii) It is known that a witness can correctly identify the cab colour only 80% of the time.

Which of the following options is closest to the probability that the accident was caused by a blue cab?

A. 12% **B.** 15% **C.** 41% **D.** 80%

Q18.5.10 MCQ 2018 2M

A coastal region with unparalleled beauty is home to many species of animals. It is dotted with coral reefs and unspoilt white sandy beaches. It has remained inaccessible to tourists due to poor connectivity and lack of accommodation. A company has spotted the opportunity and is planning to develop a luxury resort with helicopter service to the nearest major city airport. Environmentalists are upset that this would lead to the region becoming crowded and polluted like any other major beach resorts.
Which one of the following statements can be logically inferred from the information given in the above paragraph?

A. The culture and tradition of the local people will be influenced by the tourists.

B. The region will become crowded and polluted due to tourism.

C. The coral reefs are on the decline and could soon vanish.

D. Helicopter connectivity would lead to an increase in tourists coming to the region.

Session : 6 | Afternoon | 10/02/18

Q18.6.1 MCQ 2018 1M

"Since you have gone off the _____, the _____ sand is likely to damage the car."
The words that best fill the blanks in the above sentence are

A. course, coarse **B.** course, course

C. coarse, course **D.** coarse, coarse

Q18.6.2 MCQ 2018 1M

"A common misconception among writers is that sentence structure mirrors thought; the more _____ the structure, the more complicated the ideas."
The word that best fills the blank in the above sentence is

A. detailed **B.** simple **C.** clear **D.** convoluted

Q18.6.3 MCQ 2018 1M

The three roots of the equation $f(x) = 0$ are $x = \{-2, 0, 3\}$. What are the three values of x for which $f(x - 3) = 0$?

A. $-5, -3, 0$ **B.** $-2, 0, 3$ **C.** $0, 6, 8$ **D.** $1, 3, 6$

Q18.6.4 MCQ 2018 1M

For what values of k given below is $\frac{(k+2)^2}{k-3}$ an integer?

A. 4, 8, 18 **B.** 4, 10, 16 **C.** 4, 8, 28 **D.** 8, 26, 28

Q18.6.5 MCQ 2018 1M

Functions $F(a, b)$ and $G(a, b)$ are defined as follows: $F(a, b) = (a - b)^2$ and $G(a, b) = |a - b|$, where $|x|$ represents the absolute value of x.
What would be the value of $G(F(1, 3), G(1, 3))$?

A. 2 **B.** 4 **C.** 6 **D.** 36

Q18.6.6 MCQ 2018 2M

An e-mail password must contain three characters. The password has to contain one numeral from 0 to 9, one upper case and one lower case character from the English alphabet. How many distinct passwords are possible?

A. 6,760 **B.** 13,520 **C.** 40,560 **D.** 1,05,456

Q18.6.7 MCQ 2018 2M

In a certain code, AMCF is written as EQGJ and NKUF is written as ROYJ. How will DHLP be written in that code?

A. RSTN **B.** TLPH **C.** HLPT **D.** XSVR

Q18.6.8 MCQ 2018 2M

A class of twelve children has two more boys than girls. A group of three children are randomly picked from this class to accompany the teacher on a field trip. What is the probability that the group accompanying the teacher contains more girls than boys?

A. 0 **B.** $\frac{325}{864}$ **C.** $\frac{525}{864}$ **D.** $\frac{5}{12}$

Q18.6.9 MCQ 2018 2M

A designer uses marbles of four different colours for his designs. The cost of each marble is the same, irrespective of the colour. The table below shows the percentage of marbles of each colour used in the current design. The cost of each marble increased by 25%. Therefore, the designer decided to reduce equal numbers of marbles of each colour to keep the total cost unchanged. What is the percentage of blue marbles in the new design?

Blue	Black	Red	Yellow
40%	25%	20%	15%

A. 35.75 **B.** 40.25 **C.** 43.75 **D.** 46.25

Q18.6.10 MCQ 2018 2M

P, Q, R and S crossed a lake in a boat that can hold a maximum of two persons, with only one set of oars. The following additional facts are available.

(i) The boat held two persons on each of the three forward trips across the lake and one person on each of the two return trips.
(ii) P is unable to row when someone else is in the boat.
(iii) Q is unable to row with anyone else except R.
(iv) Each person rowed for at least one trip.
(v) Only one person can row during a trip.

Who rowed twice?

A. P **B.** Q **C.** R **D.** S

Session : 7 | Forenoon | 11/02/18

Q18.7.1 MCQ 2018 1M

"The driver applied the _____ as soon as she approached the hotel where she wanted to take a _____."
The words that best fill the blanks in the above sentence are

A. brake, break **B.** break, break

C. brake, brake **D.** break, brake

Q18.7.2 MCQ 2018 1M

"It is no surprise that every society has had codes of behaviour; however, the nature of these codes is often _____."
The word that best fills the blank in the above sentence is

A. unpredictable **B.** simple **C.** expected **D.** strict

Q18.7.3 MCQ 2018 1M

Hema's age is 5 years more than twice Hari's age. Suresh's age is 13 years less than 10 times Hari's age. If Suresh is 3 times as old as Hema, how old is Hema?

A. 14 **B.** 17 **C.** 18 **D.** 19

Q18.7.4 MCQ 2018 1M

Tower A is 90 m tall and tower B is 140 m tall. They are 100 m apart. A horizontal skywalk connects the floors at 70 m in both the towers. If a taut rope connects the top of tower A to the bottom of tower B, at what distance (in metres) from tower A will the rope intersect the skywalk?

A. 22.22 **B.** 50 **C.** 57.87 **D.** 77.78

Q18.7.5 MCQ 2018 1M

The temperature T in a room varies as a function of the outside temperature T_0 and the number of persons in the room p, according to the relation $T = K(\Theta p + T_0)$, where Θ and K are constants. What would be the value of Θ given the following data?

T_0	p	T
25	2	32.4
30	5	42.0

A. 0.8 **B.** 1.0 **C.** 2.0 **D.** 10.0

Q18.7.6 MCQ 2018 2M

A fruit seller sold a basket of fruits at 12.5% loss. Had he sold it for Rs. 108 more, he would have made a 10% gain. What is the loss in Rupees incurred by the fruit seller?

A. 48 **B.** 52 **C.** 60 **D.** 108

Q18.7.7 MCQ 2018 2M

The price of a wire made of a superalloy material is proportional to the square of its length. The price of 10 m length of the wire is Rs. 1600. What would be the total price (in Rs.) of two wires of lengths 4 m and 6 m?

A. 768 **B.** 832 **C.** 1440 **D.** 1600

Q18.7.8 MCQ 2018 2M

Which of the following function(s) is an accurate description of the graph for the range(s) indicated?

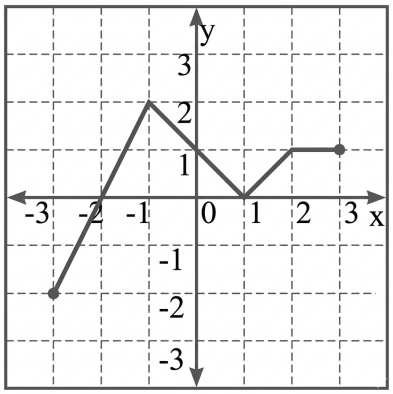


Fig. Q18.7.8

(i) $y = 2x + 4$ for $-3 \le x \le -1$
(ii) $y = |x - 1|$ for $-1 \le x \le 2$
(iii) $y = \big||x| - 1\big|$ for $-1 \le x \le 2$
(iv) $y = 1$ for $2 \le x \le 3$

A. (i), (ii) and (iii) only **B.** (i), (ii) and (iv) only

C. (i) and (iv) only **D.** (ii) and (iv) only

Q18.7.9 MCQ 2018 2M

Consider a sequence of numbers $a_1, a_2, a_3, \ldots, a_n$ where

$$a_n = \frac{1}{n} - \frac{1}{n+2}, \quad \text{for each integer } n > 0.$$

What is the sum of the first 50 terms?

A. $\left(1 + \frac{1}{2}\right) - \frac{1}{50}$ **B.** $\left(1 + \frac{1}{2}\right) + \frac{1}{50}$

C. $\left(1 + \frac{1}{2}\right) - \left(\frac{1}{51} + \frac{1}{52}\right)$ **D.** $1 - \left(\frac{1}{51} + \frac{1}{52}\right)$

Q18.7.10 MCQ 2018 2M

Each of the letters arranged as below represents a unique integer from 1 to 9. The letters are positioned in the figure such that $(A \times B \times C)$, $(B \times G \times E)$ and $(D \times E \times F)$ are equal. Which integer among the following choices cannot be represented by the letters A, B, C, D, E, F or G?

A		D
B	G	E
C		F

Fig. Q18.7.10

A. 4 **B.** 5 **C.** 6 **D.** 9

Session : 8 | Afternoon | 11/02/18

Q18.8.1 MCQ 2018 1M

"His face _____ with joy when the solution of the puzzle was _____ to him."
The words that best fill the blanks in the above sentence are

A. shone, shown **B.** shone, shone

C. shown, shone **D.** shown, shown

Q18.8.2 MCQ 2018 1M

"Although it does contain some pioneering ideas, one would hardly characterize the work as _____."
The word that best fills the blank in the above sentence is

A. innovative **B.** simple **C.** dull **D.** boring

Q18.8.3 MCQ 2018 1M

$\underbrace{a + a + a + \cdots + a}_{n \text{ times}} = a^2 b$ and $\underbrace{b + b + b + \cdots + b}_{m \text{ times}} = ab^2$,

where a, b, n and m are natural numbers.
What is the value of

$$\left(\underbrace{m + m + m + \cdots + m}_{n \text{ times}}\right)\left(\underbrace{n + n + n + \cdots + n}_{m \text{ times}}\right)?$$

A. $2a^2b^2$ **B.** a^4b^4 **C.** $ab(a + b)$ **D.** $a^2 + b^2$

Q18.8.4 MCQ 2018 1M

A three-member committee has to be formed from a group of 9 people. How many such distinct committees can be formed?

A. 27 **B.** 72 **C.** 81 **D.** 84

Q18.8.5 MCQ 2018 1M

For non-negative integers, a, b, c, what would be the value of $a + b + c$ if

$$\log a + \log b + \log c = 0?$$

A. 3 **B.** 1 **C.** 0 **D.** -1

Q18.8.6 MCQ 2018 2M

In manufacturing industries, loss is usually taken to be proportional to the square of the deviation from a target. If the loss is Rs. 4900 for a deviation of 7 units, what would be the loss in Rupees for a deviation of 4 units from the target?

A. 400 **B.** 1200 **C.** 1600 **D.** 2800

Q18.8.7 MCQ 2018 2M

A faulty wall clock is known to gain 15 minutes every 24 hours. It is synchronized to the correct time at 9 AM on 11th July. What will be the correct time to the nearest minute when the clock shows 2 PM on 15th July of the same year?

A. 12:45 PM **B.** 12:58 PM **C.** 1:00 PM **D.** 2:00 PM

Q18.8.8 MCQ 2018 2M

The annual average rainfall in a tropical city is 1000 mm. On a particular rainy day (24-hour period), the cumulative rainfall experienced by the city is shown in the graph. Over the 24-hour period, 50% of the rainfall falling on a rooftop, which had an obstruction-free area of 50 m^2, was harvested into a tank. What is the total volume of water collected in the tank in liters?

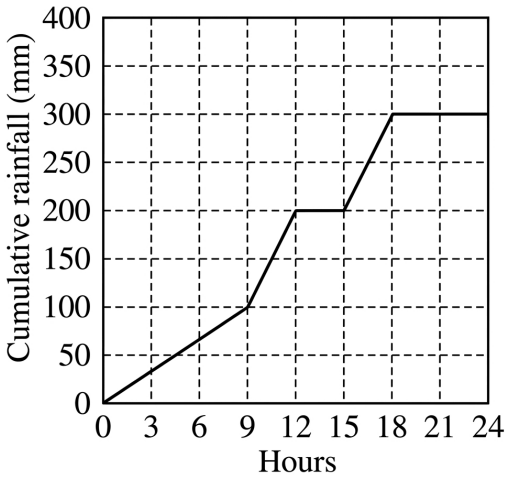


Fig. Q18.8.8

A. 25,000 **B.** 18,750 **C.** 7,500 **D.** 3,125

Q18.8.9 MCQ 2018 2M

Given that $\frac{\log P}{y-z} = \frac{\log Q}{z-x} = \frac{\log R}{x-y} = 10$ for $x \neq y \neq z$, what is the value of the product PQR?

A. 0 **B.** 1 **C.** xyz **D.** 10^{xyz}

Q18.8.10 MCQ 2018 2M

Each of the letters in the figure below represents a unique integer from 1 to 9. The letters are positioned in the figure such that each of (A+B+C), (C+D+E), (E+F+G) and (G+H+K) is equal to 13. Which integer does E represent?

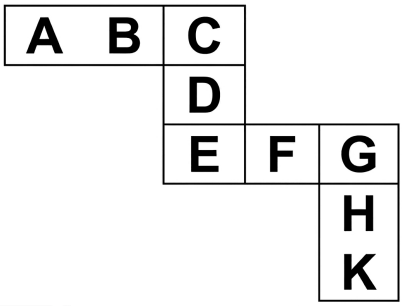


Fig. Q18.8.10

A. 1 **B.** 4 **C.** 6 **D.** 7

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q18.1.1 A	Q18.1.2 C	Q18.1.3 B	Q18.1.4 B	Q18.1.5 B	Q18.1.6 A
Q18.1.7 D	Q18.1.8 D	Q18.1.9 B	Q18.1.10 C		
Session : 2					
Q18.2.1 D	Q18.2.2 A	Q18.2.3 B	Q18.2.4 A	Q18.2.5 C	Q18.2.6 C
Q18.2.7 C	Q18.2.8 MTA	Q18.2.9 D	Q18.2.10 B		
Session : 3					
Q18.3.1 B	Q18.3.2 A	Q18.3.3 D	Q18.3.4 C	Q18.3.5 B	Q18.3.6 B
Q18.3.7 C	Q18.3.8 B	Q18.3.9 A	Q18.3.10 C		
Session : 4					
Q18.4.1 A	Q18.4.2 C	Q18.4.3 B	Q18.4.4 C	Q18.4.5 D	Q18.4.6 A
Q18.4.7 C	Q18.4.8 D	Q18.4.9 C	Q18.4.10 A		
Session : 5					
Q18.5.1 B	Q18.5.2 A	Q18.5.3 B	Q18.5.4 D	Q18.5.5 B	Q18.5.6 B
Q18.5.7 B	Q18.5.8 C	Q18.5.9 C	Q18.5.10 D		
Session : 6					
Q18.6.1 A	Q18.6.2 D	Q18.6.3 D	Q18.6.4 C	Q18.6.5 A	Q18.6.6 C
Q18.6.7 C	Q18.6.8 MTA	Q18.6.9 C	Q18.6.10 C		
Session : 7					
Q18.7.1 A	Q18.7.2 A	Q18.7.3 D	Q18.7.4 MTA	Q18.7.5 B	Q18.7.6 C
Q18.7.7 B	Q18.7.8 B	Q18.7.9 C	Q18.7.10 B		
Session : 8					
Q18.8.1 A	Q18.8.2 A	Q18.8.3 B	Q18.8.4 D	Q18.8.5 A	Q18.8.6 C
Q18.8.7 B	Q18.8.8 C	Q18.8.9 B	Q18.8.10 B		

YEAR

2017

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	ME-1, AR, TF	10
Session 2	ME-2, AG, GG	10
Session 3	EC-1, BT, CY, EY, PI	10
Session 4	EC-2, AE, MA, MT, PE, PH	10
Session 5	CS-1, EE-1	10
Session 6	CS-2, EE-2	10
Session 7	CE-1, CH, MN	10
Session 8	CE-2, IN, XE, XL	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 04/02/17

Q17.1.1 MCQ 2017 1M

He was one of my best _____ and I felt his loss _____.

A. friend, keenly **B.** friends, keen

C. friend, keener **D.** friends, keenly

Q17.1.2 MCQ 2017 1M

As the two speakers became increasingly agitated, the debate became _____.

A. lukewarm **B.** poetic **C.** forgiving **D.** heated

Q17.1.3 MCQ 2017 1M

A right-angled cone (with base radius 5 cm and height 12 cm), as shown in the figure below, is rolled on the ground keeping the point P fixed until the point Q (at the base of the cone, as shown) touches the ground again.

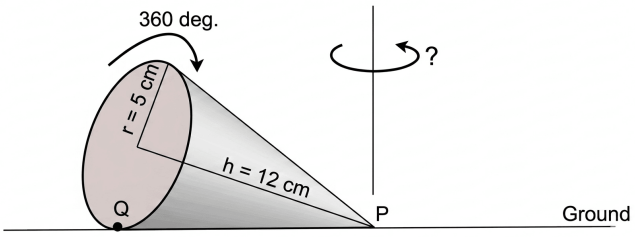


Fig. Q17.1.3

By what angle (in radians) does the cone travel?

A. $\frac{5\pi}{12}$ **B.** $\frac{5\pi}{24}$ **C.** $\frac{24\pi}{5}$ **D.** $\frac{10\pi}{13}$

Q17.1.4 MCQ 2017 1M

In a company with 100 employees, 45 earn Rs. 20,000 per month, 25 earn Rs. 30,000, 20 earn Rs. 40,000, 8 earn Rs. 60,000, and 2 earn Rs. 1,50,000. The median of the salaries is

A. Rs. 20,000 **B.** Rs. 30,000

C. Rs. 32,300 **D.** Rs. 40,000

Q17.1.5 MCQ 2017 1M

P, Q, and R talk about S's car collection. P states that S has at least 3 cars. Q believes that S has less than 3 cars. R indicates that to his knowledge, S has at least one car. Only one of P, Q and R is right. The number of cars owned by S is

A. 0 **B.** 1 **C.** 3 **D.** Cannot be determined

Q17.1.6 MCQ 2017 2M

"Here, throughout the early 1820s, Stuart continued to fight his losing battle to allow his sepoys to wear their caste-marks and their own choice of facial hair on parade, being again reprimanded by the commander-in-chief. He retort that 'A stronger instance than this of European prejudice with relation to this country has never come under my observations' had no effect on his superiors."

According to this paragraph, which of the statements below is most accurate?

A. Stuart's commander-in-chief was moved by this demonstration of his prejudice.

B. The Europeans were accommodating of the sepoys' desire to wear their caste-marks.

C. Stuart's 'losing battle' refers to his inability to succeed in enabling sepoys to wear caste-marks.

D. The commander-in-chief was exempt from the European prejudice that dictated how the sepoys were to dress.

Q17.1.7 MCQ 2017 2M

What is the sum of the missing digits in the subtraction problem below?

$$\begin{array}{r} 5____ \\ -\,48_89 \\ \hline 1111 \end{array}$$

A. 8 **B.** 10 **C.** 11 **D.** Cannot be determined

Q17.1.8 MCQ 2017 2M

Let S_1 be the plane figure consisting of the points (x, y) given by the inequalities $|x - 1| \leq 2$ and $|y + 2| \leq 3$. Let S_2 be the plane figure given by the inequalities $x - y \geq -2$, $y \geq 1$, and $x \leq 3$. Let S be the union of S_1 and S_2. The area of S is

A. 26 **B.** 28 **C.** 32 **D.** 34

Q17.1.9 MCQ 2017 2M

Two very famous sportsmen Mark and Steve happened to be brothers, and played for country K. Mark teased James, an opponent from country E, "There is no way you are good enough to play for your country." James replied, "Maybe not, but at least I am the best player in my own family."

Which one of the following can be inferred from this conversation?

A. Mark was known to play better than James

B. Steve was known to play better than Mark

C. James and Steve were good friends

D. James played better than Steve

Q17.1.10 MCQ 2017 2M

The growth of bacteria (lactobacillus) in milk leads to curd formation. A minimum bacterial population density of 0.8 (in suitable units) is needed to form curd. In the graph below, the population density of lactobacillus in 1 litre of milk is plotted as a function of time, at two different temperatures, 25 °C and 37 °C.

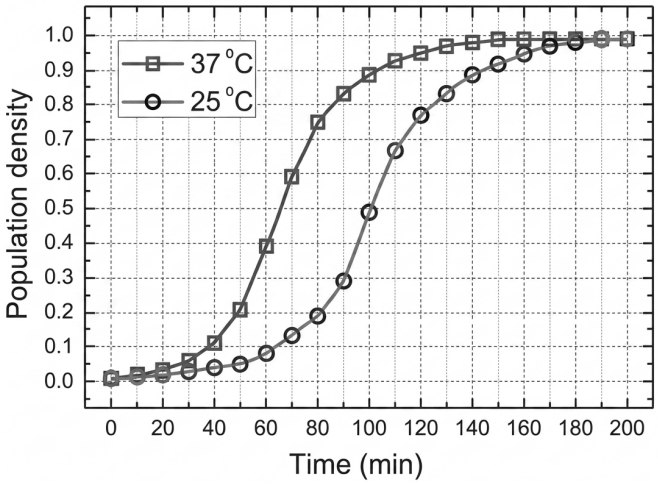


Fig. Q17.1.10

Consider the following statements based on the data shown above:

i. The growth in bacterial population stops earlier at 37 °C as compared to 25 °C

ii. The time taken for curd formation at 25 °C is twice the time taken at 37 °C

Which one of the following options is correct?

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Session : 2 | Afternoon | 04/02/17

Q17.2.1 MCQ 2017 1M

The ways in which this game can be played _____ potentially infinite.

A. is **B.** is being **C.** are **D.** are being

Q17.2.2 MCQ 2017 1M

If you choose plan P, you will have to _____ plan Q, as these two are mutually _____.

A. forgo, exclusive **B.** forget, inclusive

C. accept, exhaustive **D.** adopt, intrusive

Q17.2.3 MCQ 2017 1M

If a and b are integers and $a - b$ is even, which of the following must always be even?

A. ab **B.** $a^2 + b^2 + 1$ **C.** $a^2 + b + 1$ **D.** $ab - b$

Q17.2.4 MCQ 2017 1M

A couple has 2 children. The probability that both children are boys if the older one is a boy is

A. 1/4 **B.** 1/3 **C.** 1/2 **D.** 1

Q17.2.5 MCQ 2017 1M

P looks at Q while Q looks at R. P is married, R is not. The number of pairs of people in which a married person is looking at an unmarried person is

A. 0 **B.** 1 **C.** 2 **D.** Cannot be determined

Q17.2.6 MCQ 2017 2M

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters."
Which of the following is closest in meaning to 'cleaving'?

A. deteriorating **B.** arguing **C.** departing **D.** splitting

Q17.2.7 MCQ 2017 2M

X bullocks and Y tractors take 8 days to plough a field. If we halve the number of bullocks and double the number of tractors, it takes 5 days to plough the same field. How many days will it take X bullocks alone to plough the field?

A. 30 **B.** 35 **C.** 40 **D.** 45

Q17.2.8 MCQ 2017 2M

There are 4 women P, Q, R, S, and 5 men V, W, X, Y, Z in a group. We are required to form pairs each consisting of one woman and one man. P is not to be paired with Z, and Y must necessarily be paired with someone. In how many ways can 4 such pairs be formed?

A. 74 **B.** 76 **C.** 78 **D.** 80

Q17.2.9 MCQ 2017 2M

All people in a certain island are either 'Knights' or 'Knaves' and each person knows every other person's identity.
Knights NEVER lie, and knaves ALWAYS lie.
P says "Both of us are knights".
Q says "None of us are knaves".
Which one of the following can be logically inferred from the above?

A. Both P and Q are knights

B. P is a knight; Q is a knave

C. Both P and Q are knaves

D. The identities of P, Q cannot be determined

Q17.2.10 MCQ 2017 2M

In the graph below, the concentration of a particular pollutant in a lake is plotted over (alternate) days of a month in winter (average temperature 10 °C) and a month in summer (average temperature 30 °C).

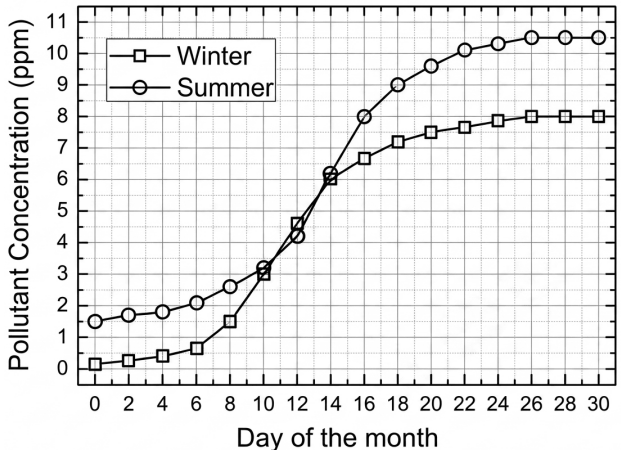


Fig. Q17.2.10

Consider the following statements based on the data shown above:

i. Over the given months, the difference between the maximum and the minimum pollutant concentrations is the same in both winter and summer.
ii. There are at least four days in the summer month such that the pollutant concentrations on those days are within 1 ppm of the pollutant concentrations on the corresponding days in the winter month.

Which one of the following options is correct?

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Session : 3 | Forenoon | 05/02/17

Q17.3.1 MCQ 2017 1M

She has a sharp tongue and it can occasionally turn___

A. hurtful **B.** left **C.** methodical **D.** vital

Q17.3.2 MCQ 2017 1M

I _____ made arrangements had I _____ informed earlier.

A. could have, been **B.** would have, being

C. had, have **D.** had been, been

Q17.3.3 MCQ 2017 1M

In the summer, water consumption is known to decrease overall by 25%. A Water Board official states that in the summer household consumption decreases by 20%, while other consumption increases by 70%.
Which of the following statements is correct?

A. The ratio of household to other consumption is 8/17

B. The ratio of household to other consumption is 1/17

C. The ratio of household to other consumption is 17/8

D. There are errors in the official's statement.

Q17.3.4 MCQ 2017 1M

40% of deaths on city roads may be attributed to drunken driving. The number of degrees needed to represent this as a slice of a pie chart is

A. 120 **B.** 144 **C.** 160 **D.** 212

Q17.3.5 MCQ 2017 1M

Some tables are shelves. Some shelves are chairs. All chairs are benches. Which of the following conclusions can be deduced from the preceding sentences?

i. At least one bench is a table
ii. At least one shelf is a bench
iii. At least one chair is a table
iv. All benches are chairs

A. Only i **B.** Only ii **C.** Only ii and iii **D.** Only iv

Q17.3.6 MCQ 2017 2M

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters".
Here, the word 'antagonistic' is closest in meaning to

A. impartial **B.** argumentative **C.** separated **D.** hostile

Q17.3.7 MCQ 2017 2M

S, T, U, V, W, X, Y, and Z are seated around a circular table. T's neighbours are Y and V. Z is seated third to the left of T and second to the right of S. U's neighbours are S and Y, and T and W are not seated opposite each other. Who is third to the left of V?

A. X **B.** W **C.** U **D.** T

Q17.3.8 MCQ 2017 2M

Trucks (10 m long) and cars (5 m long) go on a single lane bridge. There must be a gap of at least 20 m after each truck and a gap of at least 15 m after each car. Trucks and cars travel at a speed of 36 km/h. If cars and trucks go alternately, what is the maximum number of vehicles that can use the bridge in one hour?

A. 1440 **B.** 1200 **C.** 720 **D.** 600

Q17.3.9 MCQ 2017 2M

There are 3 Indians and 3 Chinese in a group of 6 people. How many subgroups of this group can we choose so that every subgroup has at least one Indian?

A. 56 **B.** 52 **C.** 48 **D.** 44

Q17.3.10 MCQ 2017 2M

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.

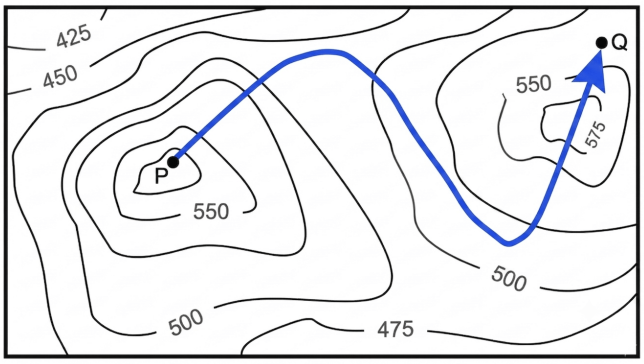


Fig. Q17.3.10

The path from P to Q is best described by

A. Up-Down-Up-Down **B.** Down-Up-Down-Up

C. Down-Up-Down **D.** Up-Down-Up

Session : 4 | Afternoon | 05/02/17

Q17.4.1 MCQ 2017 1M

The ninth and the tenth of this month are Monday and Tuesday _____.

A. figuratively **B.** retrospectively

C. respectively **D.** rightfully

Q17.4.2 MCQ 2017 1M

It is _____ to read this year's textbook _____ the last year's.

A. easier, than **B.** most easy, than

C. easier, from **D.** easiest, from

Q17.4.3 MCQ 2017 1M

A rule states that in order to drink beer, one must be over 18 years old. In a bar, there are 4 people. P is 16 years old, Q is 25 years old, R is drinking milkshake and S is drinking a beer. What must be checked to ensure that the rule is being followed?

A. Only P's drink

B. Only P's drink and S's age

C. Only S's age

D. Only P's drink, Q's drink and S's age

Q17.4.4 MCQ 2017 1M

Fatima starts from point P, goes North for 3 km, and then East for 4 km to reach point Q. She then turns to face point P and goes 15 km in that direction. She then goes North for 6 km. How far is she from point P, and in which direction should she go to reach point P?

A. 8 km, East **B.** 12 km, North

C. 6 km, East **D.** 10 km, North

Q17.4.5 MCQ 2017 1M

500 students are taking one or more courses out of Chemistry, Physics, and Mathematics. Registration records indicate course enrolment as follows: Chemistry (329), Physics (186), Mathematics (295), Chemistry and Physics (83), Chemistry and Mathematics (217), and Physics and Mathematics (63). How many students are taking all 3 subjects?

A. 37 **B.** 43 **C.** 47 **D.** 53

Q17.4.6 MCQ 2017 2M

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters."
Which of the following statements best reflects the author's opinion?

A. An intimate association does not allow for the necessary perspective.

B. Matters are recorded with an impartial perspective.

C. An intimate association offers an impartial perspective.

D. Actors are typically associated with the impartial recording of matters.

Q17.4.7 MCQ 2017 2M

Each of P, Q, R, S, W, X, Y and Z has been married at most once. X and Y are married and have two children P and Q. Z is the grandfather of the daughter S of P. Further, Z and W are married and are parents of R.
Which one of the following must necessarily be FALSE?

A. X is the mother-in-law of R

B. P and R are not married to each other

C. P is a son of X and Y

D. Q cannot be married to R

Q17.4.8 MCQ 2017 2M

1200 men and 500 women can build a bridge in 2 weeks. 900 men and 250 women will take 3 weeks to build the same bridge. How many men will be needed to build the bridge in one week?

A. 3000 **B.** 3300 **C.** 3600 **D.** 3900

Q17.4.9 MCQ 2017 2M

The number of 3-digit numbers such that the digit 1 is never to the immediate right of 2 is

A. 781 **B.** 791 **C.** 881 **D.** 891

Q17.4.10 MCQ 2017 2M

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.

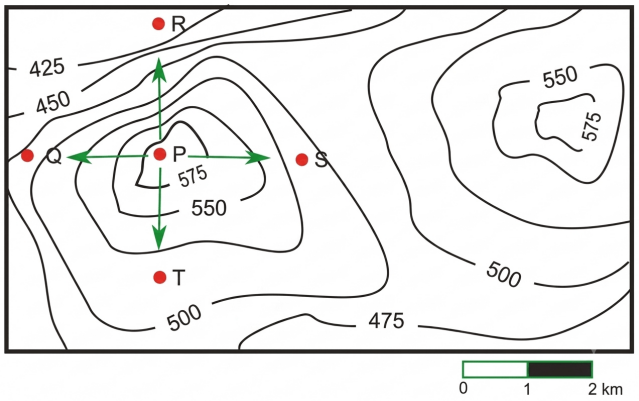


Fig. Q17.4.10

Which of the following is the steepest path leaving from P?

A. P to Q **B.** P to R **C.** P to S **D.** P to T

Session : 5 | Forenoon | 11/02/17

Q17.5.1 MCQ 2017 1M

After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

A. was wishing **B.** is wishing **C.** wished **D.** had wished

Q17.5.2 MCQ 2017 1M

Research in the workplace reveals that people work for many reasons _____.

A. money beside **B.** beside money

C. money besides **D.** besides money

Q17.5.3 MCQ 2017 1M

Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

A. Rahul and Murali **B.** Srinivas and Arul

C. Srinivas and Murali **D.** Srinivas and Rahul

Q17.5.4 MCQ 2017 1M

Find the smallest number y such that $y \times 162$ is a perfect cube.

A. 24 **B.** 27 **C.** 32 **D.** 36

Q17.5.5 MCQ 2017 1M

The probability that a k-digit number does NOT contain the digits 0, 5, or 9 is

A. 0.3^k **B.** 0.6^k **C.** 0.7^k **D.** 0.9^k

Q17.5.6 MCQ 2017 2M

"The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history." Which of the following statements best reflects the author's opinion?

A. Nationalists are highly imaginative.

B. History is viewed through the filter of nationalism.

C. Our colonial past never happened.

D. Nationalism has to be both adequately and properly imagined.

Q17.5.7 MCQ 2017 2M

Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

A. 2 **B.** 3 **C.** 4 **D.** Cannot be determined

Q17.5.8 MCQ 2017 2M

The expression $\dfrac{(x+y)-|x-y|}{2}$ is equal to

A. the maximum of x and y **B.** the minimum of x and y

C. 1 **D.** none of the above

Q17.5.9 MCQ 2017 2M

Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

A. 21 **B.** 18 **C.** 16 **D.** 14

Q17.5.10 MCQ 2017 2M

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?

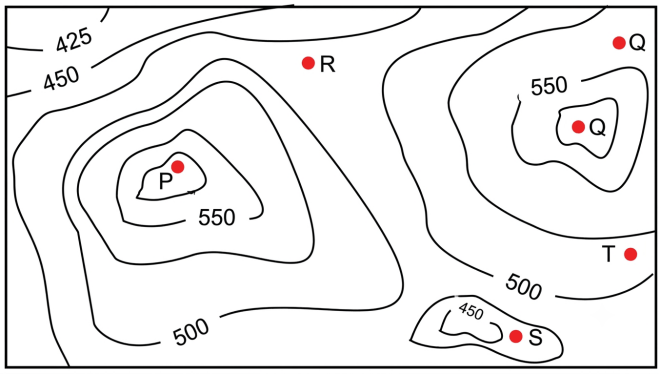


Fig. Q17.5.10

A. P, Q **B.** P, Q, T **C.** R, S, T **D.** Q, R, S

Session : 6 | Afternoon | 11/02/17

Q17.6.1 MCQ 2017 1M

Choose the option with words that are not synonyms.

A. aversion, dislike **B.** luminous, radiant

C. plunder, loot **D.** yielding, resistant

Q17.6.2 MCQ 2017 1M

Saturn is _____ to be seen on a clear night with the naked eye.

A. enough bright **B.** bright enough

C. as enough bright **D.** bright as enough

Q17.6.3 MCQ 2017 1M

There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?

A. V **B.** W **C.** X **D.** Y

Q17.6.4 MCQ 2017 1M

A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

A. 12 **B.** 15 **C.** 18 **D.** 19

Q17.6.5 MCQ 2017 1M

There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

A. 1/5 **B.** 7/30 **C.** 1/4 **D.** 4/15

Q17.6.6 MCQ 2017 2M

"We lived in a culture that denied any merit to literary works, considering them important only when they were handmaidens to something seemingly more urgent – namely ideology. This was a country where all gestures, even the most private, were interpreted in political terms."
The author's belief that ideology is not as important as literature is revealed by the word:

A. 'culture' **B.** 'seemingly'

C. 'urgent' **D.** 'political'

Q17.6.7 MCQ 2017 2M

There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labelled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

A. The box labelled 'Apples'

B. The box labelled 'Apples and Oranges'

C. The box labelled 'Oranges'

D. Cannot be determined

Q17.6.8 MCQ 2017 2M

X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

A. 90 digits **B.** 91 digits **C.** 92 digits **D.** 93 digits

Q17.6.9 MCQ 2017 2M

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q17.6.10 MCQ 2017 2M

An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.

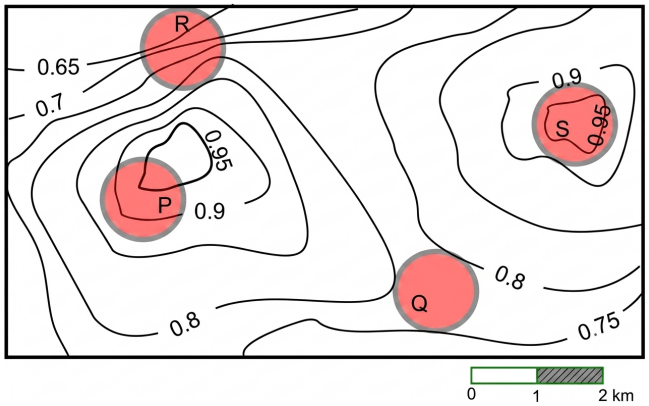


Fig. Q17.6.10

If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?

A. P **B.** Q **C.** R **D.** S

Session : 7 | Forenoon | 12/02/17

Q17.7.1 MCQ 2017 1M

The bacteria in milk are destroyed when it _____ heated to 80 degree Celsius.

A. would be **B.** will be **C.** is **D.** was

Q17.7.2 MCQ 2017 1M

_____ with someone else's email account is now a very serious offence.

A. Involving **B.** Assisting **C.** Tampering **D.** Incubating

Q17.7.3 MCQ 2017 1M

Consider the following sentences:
All benches are beds. No bed is a bulb. Some bulbs are lamps.
Which of the following can be inferred?

i. Some beds are lamps.
ii. Some lamps are beds.

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Q17.7.4 MCQ 2017 1M

If the radius of a right circular cone is increased by 50%, its volume increases by

A. 75% **B.** 100% **C.** 125% **D.** 237.5%

Q17.7.5 MCQ 2017 1M

The following sequence of numbers is arranged in increasing order: 1, x, x, x, y, y, 9, 16, 18. Given that the mean and median are equal, and are also equal to twice the mode, the value of y is

A. 5 **B.** 6 **C.** 7 **D.** 8

Q17.7.6 MCQ 2017 2M

The old concert hall was demolished because of fears that the foundation would be affected by the construction of the new metro line in the area. Modern technology for underground metro construction tried to mitigate the impact of pressurized air pockets created by the excavation of large amounts of soil. But even with these safeguards, it was feared that the soil below the concert hall would not be stable.
From this, one can infer that

A. the foundations of old buildings create pressurized air pockets underground, which are difficult to handle during metro construction.

B. metro construction has to be done carefully considering its impact on the foundations of existing buildings.

C. old buildings in an area form an impossible hurdle to metro construction in that area.

D. pressurized air can be used to excavate large amounts of soil from underground areas.

Q17.7.7 MCQ 2017 2M

Students applying for hostel rooms are allotted rooms in order of seniority. Students already staying in a room will move if they get a room in their preferred list. Preferences of lower ranked applicants are ignored during allocation. Given the data below, which room will Ajit stay in?

Names	Student seniority	Current room	Room preference list
Amar	1	P	R, S, Q
Akbar	2	None	R, S
Anthony	3	Q	P
Ajit	4	S	Q, P, R

A. P **B.** Q **C.** R **D.** S

Q17.7.8 MCQ 2017 2M

The last digit of $(2171)^7 + (2172)^9 + (2173)^{11} + (2174)^{13}$ is

A. 2 **B.** 4 **C.** 6 **D.** 8

Q17.7.9 MCQ 2017 2M

Two machines M1 and M2 are able to execute any of four jobs P, Q, R and S. The machines can perform one job on one object at a time. Jobs P, Q, R and S take 30 minutes, 20 minutes, 60 minutes and 15 minutes each respectively. There are 10 objects each requiring exactly 1 job. Job P is to be performed on 2 objects, Job Q on 3 objects, Job R on 1 object and Job S on 4 objects. What is the minimum time needed to complete all the jobs?

A. 2 hours **B.** 2.5 hours **C.** 3 hours **D.** 3.5 hours

Q17.7.10 MCQ 2017 2M

The bar graph below shows the output of five carpenters over one month, each of whom made different items of furniture: chairs, tables, and beds.

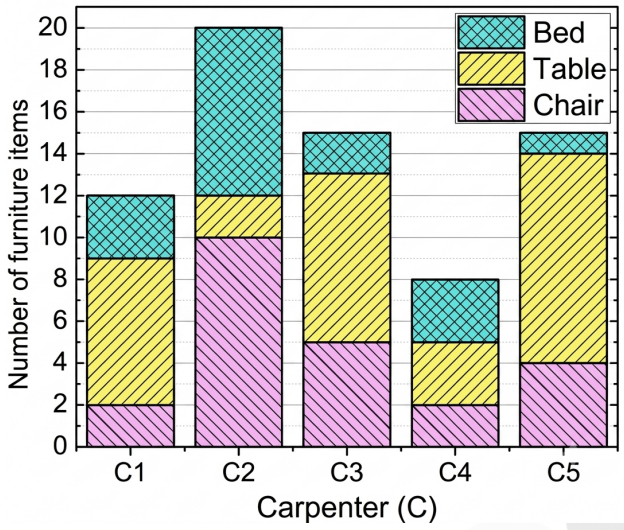


Fig. Q17.7.10

Consider the following statements:

i. The number of beds made by carpenter C2 is exactly the same as the number of tables made by carpenter C3.

ii. The total number of chairs made by all carpenters is less than the total number of tables.

Which one of the following is true?

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Session : 8 | Afternoon | 12/02/17

Q17.8.1 MCQ 2017 1M

The event would have been successful if you _____ able to come.

A. are **B.** had been **C.** have been **D.** would have been

Q17.8.2 MCQ 2017 1M

There was no doubt that their work was thorough.

Which of the words below is closest in meaning to the underlined word above?

A. pretty **B.** complete **C.** sloppy **D.** haphazard

Q17.8.3 MCQ 2017 1M

Four cards lie on a table. Each card has a number printed on one side and a colour on the other. The faces visible on the cards are 2, 3, red, and blue.

Proposition: If a card has an even value on one side, then its opposite face is red.

The cards which MUST be turned over to verify the above proposition are

A. 2, red **B.** 2, 3, red **C.** 2, blue **D.** 2, red, blue

Q17.8.4 MCQ 2017 1M

What is the value of x when $81\times\left(\frac{16}{25}\right)^{x+2}\div\left(\frac{2}{5}\right)^{2x+4}=144$?

A. 1 **B.** -1 **C.** -2 **D.** Cannot be determined

Q17.8.5 MCQ 2017 1M

Two dice are thrown simultaneously. The probability that the product of the numbers appearing on the top faces of the dice is a perfect square is

A. 1/9 **B.** 2/9 **C.** 1/3 **D.** 4/9

Q17.8.6 MCQ 2017 2M

Bhaichung was observing the pattern of people entering and leaving a car service centre. There was a single window where customers were being served. He saw that people inevitably came out of the centre in the order that they went in. However, the time they spent inside seemed to vary a lot: some people came out in a matter of minutes while for others it took much longer.

From this, what can one conclude?

A. The centre operates on a first-come-first-served basis, but with variable service times depending on specific customer needs.

B. Customers were served in an arbitrary order, since they took varying amounts of time for service completion in the centre.

C. Since some people came out within a few minutes of entering the centre, the system is likely to operate on a last-come-first-served basis.

D. Entering the centre early ensured that one would have shorter service times and most people attempted to do this.

Q17.8.7 MCQ 2017 2M

A map shows the elevations of Darjeeling, Gangtok, Kalimpong, Pelling, and Siliguri. Kalimpong is at a lower elevation than Gangtok. Pelling is at a lower elevation than Gangtok. Pelling is at a higher elevation than Siliguri. Darjeeling is at a higher elevation than Gangtok.
Which of the following statements can be inferred from the paragraph above?

i. Pelling is at a higher elevation than Kalimpong
ii. Kalimpong is at a lower elevation than Darjeeling
iii. Kalimpong is at a lower elevation than Pelling
iv. Siliguri is at a lower elevation than Gangtok

A. Only ii
B. Only ii and iii
C. Only ii and iv
D. Only iii and iv

Q17.8.8 MCQ 2017 2M

P, Q, R, S, T and U are seated around a circular table. R is seated two places to the right of Q. P is seated three places to the left of R. S is seated opposite U. If P and U now switch seats, which of the following must necessarily be true?

A. P is immediately to the right of R

B. T is immediately to the left of P

C. T is immediately to the left of P or P is immediately to the right of Q

D. U is immediately to the right of R or P is immediately to the left of T

Q17.8.9 MCQ 2017 2M

Budhan covers a distance of 19 km in 2 hours by cycling one fourth of the time and walking the rest. The next day he cycles (at the same speed as before) for half the time and walks the rest (at the same speed as before) and covers 26 km in 2 hours. The speed in km/h at which Budhan walks is

A. 1 **B.** 4 **C.** 5 **D.** 6

Q17.8.10 MCQ 2017 2M

The points in the graph below represent the halts of a lift for durations of 1 minute, over a period of 1 hour.

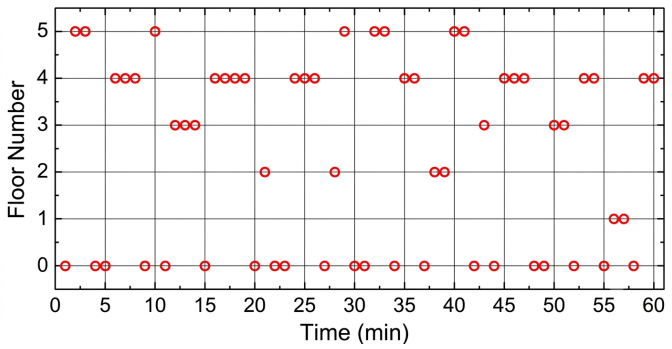


Fig. Q17.8.10

Which of the following statements are correct?

i. The elevator never moves directly from any non-ground floor to another non-ground floor over the one hour period
ii. The elevator stays on the fourth floor for the longest duration over the one hour period

A. Only i
B. Only ii
C. Both i and ii
D. Neither i nor ii

Answer Key

Session : 1					
Q17.1.1 D	Q17.1.2 D	Q17.1.3 D	Q17.1.4 B	Q17.1.5 A	Q17.1.6 C
Q17.1.7 D	Q17.1.8 C	Q17.1.9 B	Q17.1.10 A		
Session : 2					
Q17.2.1 C	Q17.2.2 A	Q17.2.3 D	Q17.2.4 C	Q17.2.5 B	Q17.2.6 D
Q17.2.7 A	Q17.2.8 C	Q17.2.9 D	Q17.2.10 B		
Session : 3					
Q17.3.1 A	Q17.3.2 A	Q17.3.3 D	Q17.3.4 B	Q17.3.5 B	Q17.3.6 D
Q17.3.7 A	Q17.3.8 A	Q17.3.9 A	Q17.3.10 C		
Session : 4					
Q17.4.1 C	Q17.4.2 A	Q17.4.3 B	Q17.4.4 A	Q17.4.5 D	Q17.4.6 A
Q17.4.7 B	Q17.4.8 C	Q17.4.9 C	Q17.4.10 B		
Session : 5					
Q17.5.1 C	Q17.5.2 D	Q17.5.3 C	Q17.5.4 D	Q17.5.5 C	Q17.5.6 B
Q17.5.7 A	Q17.5.8 B	Q17.5.9 D	Q17.5.10 C		
Session : 6					
Q17.6.1 D	Q17.6.2 B	Q17.6.3 A	Q17.6.4 B	Q17.6.5 D	Q17.6.6 B
Q17.6.7 B	Q17.6.8 A	Q17.6.9 C	Q17.6.10 C		
Session : 7					
Q17.7.1 C	Q17.7.2 C	Q17.7.3 D	Q17.7.4 C	Q17.7.5 D	Q17.7.6 B
Q17.7.7 B	Q17.7.8 B	Q17.7.9 A	Q17.7.10 C		
Session : 8					
Q17.8.1 B	Q17.8.2 B	Q17.8.3 C	Q17.8.4 B	Q17.8.5 B	Q17.8.6 A
Q17.8.7 C	Q17.8.8 C	Q17.8.9 D	Q17.8.10 D		

YEAR

2016

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	ME-1, EC-1	10
Session 2	ME-2, BT, CH, GG, MN, PH	10
Session 3	ME-3, EC-2	10
Session 4	EC-3, AR, CY, IN, MA, PE	10
Session 5	CS-1, CE-1	10
Session 6	CS-2, EE-1	10
Session 7	CE-2, AG, EY, MT, PI	10
Session 8	EE-2, AE, TF, XL , XE	10

Year Snapshot

Total Questions: 80 **MCQ:** 77 **MSQ:** 1 **NAT:** 2

Session : 1 | Forenoon | 30/01/16

Q16.1.1 MCQ 2016 1M

Which of the following is **CORRECT** with respect to grammar and usage?
Mount Everest is _____.

A. the highest peak in the world

B. highest peak in the world

C. one of highest peak in the world

D. one of the highest peak in the world

Q16.1.2 MCQ 2016 1M

The policeman asked the victim of a theft, "What did you _____?"

A. loose **B.** lose **C.** loss **D.** louse

Q16.1.3 MCQ 2016 1M

Despite the new medicine's _____ in treating diabetes, it is not _____ widely.

A. effectiveness — prescribed

B. availability — used

C. prescription — available

D. acceptance — proscribed

Q16.1.4 MCQ 2016 1M

In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?

A. 2029198 **B.** 2467482 **C.** 2789080 **D.** 3577422

Q16.1.5 MCQ 2016 1M

Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?

A. 3.00 **B.** 4.99 **C.** 6.02 **D.** 7.01

Q16.1.6 MCQ 2016 2M

A person moving through a tuberculosis prone zone has a 50% probability of becoming infected. However, only 30% of infected people develop the disease. What percentage of people moving through a tuberculosis prone zone remains infected but does not show symptoms of disease?

A. 15 **B.** 33 **C.** 35 **D.** 37

Q16.1.7 MCQ 2016 2M

In a world filled with uncertainty, he was glad to have many good friends. He had always assisted them in times of need and was confident that they would reciprocate. However, the events of the last week proved him wrong.
Which of the following inference(s) is/are logically valid and can be inferred from the above passage?

(i) His friends were always asking him to help them.
(ii) He felt that when in need of help, his friends would let him down.
(iii) He was sure that his friends would help him when in need.
(iv) His friends did not help him last week.

A. (i) and (ii) **B.** (iii) and (iv)

C. (iii) only **D.** (iv) only

Q16.1.8 MCQ 2016 2M

Leela is older than her cousin Pavithra. Pavithra's brother Shiva is older than Leela. When Pavithra and Shiva are visiting Leela, all three like to play chess. Pavithra wins more often than Leela does.
Which one of the following statements must be **TRUE** based on the above?

A. When Shiva plays chess with Leela and Pavithra, he often loses.

B. Leela is the oldest of the three.

C. Shiva is a better chess player than Pavithra.

D. Pavithra is the youngest of the three.

Q16.1.9 MCQ 2016 2M

If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $s^{-c} = \frac{1}{q}$, the value of abc is _____.

A. $(rqs)^{-1}$ **B.** 0 **C.** 1 **D.** $r + q + s$

Q16.1.10 MCQ 2016 2M

P, **Q**, **R** and **S** are working on a project. **Q** can finish the task in 25 days, working alone for 12 hours a day. **R** can finish the task in 50 days, working alone for 12 hours per day. **Q** worked 12 hours a day but took sick leave in the beginning for two days. **R** worked 18 hours a day on all days. What is the ratio of work done by **Q** and **R** after 7 days from the start of the project?

A. 10:11 **B.** 11:10 **C.** 20:21 **D.** 21:20

Session : 2 | Afternoon | 30/01/16

Q16.2.1 MCQ 2016 1M

The volume of a sphere of diameter 1 unit is _____ than the volume of a cube of side 1 unit.

A. least **B.** less **C.** lesser **D.** low

Q16.2.2 MCQ 2016 1M

The unruly crowd demanded that the accused be _____ without trial.

A. hanged **B.** hanging **C.** hankering **D.** hung

Q16.2.3 MSQ 2016 1M

Choose the statement(s) where the underlined word is used correctly:

(i) A prone is a dried plum.
(ii) He was lying prone on the floor.
(iii) People who eat a lot of fat are prone to heart disease.

A. (i) and (iii) only **B.** (iii) only

C. (i) and (ii) only **D.** (ii) and (iii) only

Q16.2.4 MCQ 2016 1M

Fact: If it rains, then the field is wet.
Read the following statements:

(i) It rains
(ii) The field is not wet
(iii) The field is wet
(iv) It did not rain

Which one of the options given below is **NOT** logically possible, based on the given fact?

A. If (iii), then (iv). **B.** If (i), then (iii).

C. If (i), then (ii). **D.** If (ii), then (iv).

Q16.2.5 MCQ 2016 1M

A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 m, the area of the window in m^2 is _____.

A. 1.43 **B.** 2.06 **C.** 2.68 **D.** 2.88

Q16.2.6 MCQ 2016 2M

Students taking an exam are divided into two groups, **P** and **Q** such that each group has the same number of students. The performance of each of the students in a test was evaluated out of 200 marks. It was observed that the mean of group **P** was 105, while that of group **Q** was 85. The standard deviation of group **P** was 25, while that of group **Q** was 5. Assuming that the marks were distributed on a normal distribution,
Which of the following statements will have the highest probability of being **TRUE**?

A. No student in group **Q** scored less marks than any student in group **P**.

B. No student in group **P** scored less marks than any student in group **Q**.

C. Most students of group **Q** scored marks in a narrower range than students in group **P**.

D. The median of the marks of group **P** is 100.

Q16.2.7 MCQ 2016 2M

A smart city integrates all modes of transport, uses clean energy and promotes sustainable use of resources. It also uses technology to ensure safety and security of the city, something which critics argue, will lead to a surveillance state.
Which of the following can be logically inferred from the above paragraph?

(i) All smart cities encourage the formation of surveillance states.
(ii) Surveillance is an integral part of a smart city.
(iii) Sustainability and surveillance go hand in hand in a smart city.
(iv) There is a perception that smart cities promote surveillance.

A. (i) and (iv) only **B.** (ii) and (iii) only

C. (iv) only **D.** (i) only

Q16.2.8 MCQ 2016 2M

Find the missing sequence in the letter series.

$B,\ FH,\ LNP,\ ____$

A. SUWY **B.** TUVW **C.** TVXZ **D.** TWXZ

Q16.2.9 MCQ 2016 2M

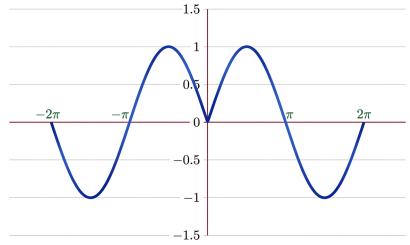
The binary operation $\square$ is defined as $a \square b = ab + (a + b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a \square x = a$, for any a, is _____.

A. 0 **B.** 1 **C.** 2 **D.** 10

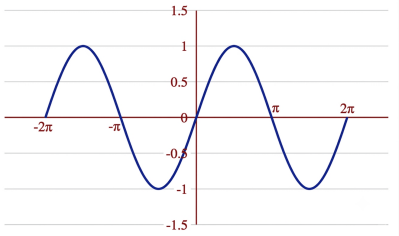
Q16.2.10 MCQ 2016 2M

Which of the following curves represents the function $y = \ln\left(\left|e^{[|\sin(|x|)|]}\right|\right)$ for $|x| < 2\pi$? Here, x represents the abscissa and y represents the ordinate.

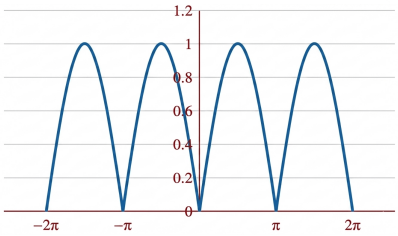
A.



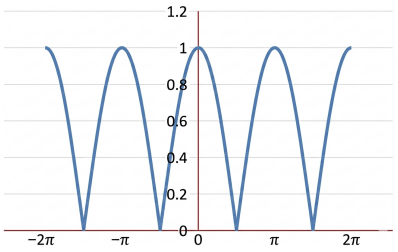
B.



C.



D.



Session : 3 | Forenoon | 31/01/16

Q16.3.1 MCQ 2016 1M

Based on the given statements, select the appropriate option with respect to grammar and usage.

Statements:

(i) The height of Mr. **X** is 6 feet.

(ii) The height of Mr. **Y** is 5 feet.

A. Mr. **X** is longer than Mr. **Y**.

B. Mr. **X** is more elongated than Mr. **Y**.

C. Mr. **X** is taller than Mr. **Y**.

D. Mr. **X** is lengthier than Mr. **Y**.

Q16.3.2 MCQ 2016 1M

The students _____ the teacher on teachers' day for twenty years of dedicated teaching.

A. facilitated **B.** felicitated **C.** fantasized **D.** facillitated

Q16.3.3 MCQ 2016 1M

After India's cricket world cup victory in 1985, Shrotria who was playing both tennis and cricket till then, decided to concentrate only on cricket. And the rest is history. What does the underlined phrase mean in this context?

A. history will rest in peace

B. rest is recorded in history books

C. rest is well known

D. rest is archaic

Q16.3.4 MCQ 2016 1M

Given $(9 \text{ inches})^{\frac{1}{2}} = (0.25 \text{ yards})^{\frac{1}{2}}$, which one of the following statements is **TRUE**?

A. 3 inches = 0.5 yards **B.** 9 inches = 1.5 yards

C. 9 inches = 0.25 yards **D.** 81 inches = 0.0625 yards

Q16.3.5 MCQ 2016 1M

S, **M**, **E** and **F** are working in shifts in a team to finish a project. **M** works with twice the efficiency of others but for half as many days as **E** worked. **S** and **M** have 6 hour shifts in a day, whereas **E** and **F** have 12 hour shifts. What is the ratio of contribution of **M** to contribution of **E** in the project?

A. 1:1 **B.** 1:2 **C.** 1:4 **D.** 2:1

Q16.3.6 MCQ 2016 2M

The Venn diagram shows the preference of the student population for leisure activities.

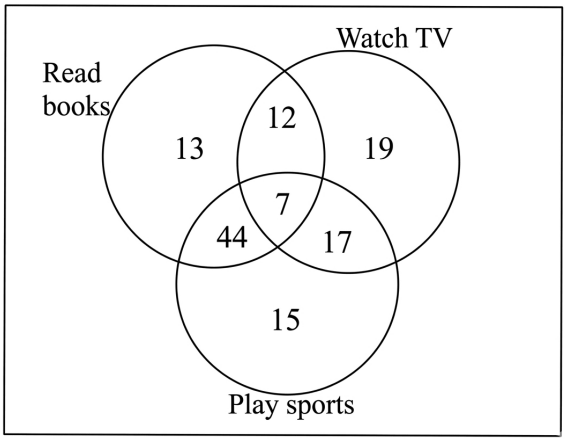


Fig. Q16.3.6

From the data given, the number of students who like to read books or play sports is _____.

A. 44 **B.** 51 **C.** 79 **D.** 108

Q16.3.7 MCQ 2016 2M

Social science disciplines were in existence in an amorphous form until the colonial period when they were institutionalized. In varying degrees, they were intended to further the colonial interest. In the time of globalization and the economic rise of postcolonial countries like India, conventional ways of knowledge production have become obsolete.

Which of the following can be logically inferred from the above statements?

(i) Social science disciplines have become obsolete.
(ii) Social science disciplines had a pre-colonial origin.
(iii) Social science disciplines always promote colonialism.
(iv) Social science must maintain disciplinary boundaries.

A. (ii) only **B.** (i) and (iii) only

C. (ii) and (iv) only **D.** (iii) and (iv) only

Q16.3.8 MCQ 2016 2M

Two and a quarter hours back, when seen in a mirror, the reflection of a wall clock without number markings seemed to show 1:30. What is the actual current time shown by the clock?

A. 8:15 **B.** 11:15 **C.** 12:15 **D.** 12:45

Q16.3.9 MCQ 2016 2M

M and **N** start from the same location. **M** travels 10 km East and then 10 km North-East. **N** travels 5 km South and then 4 km South-East. What is the shortest distance (in km) between **M** and **N** at the end of their travel?

A. 18.60 **B.** 22.50 **C.** 20.61 **D.** 25.00

Q16.3.10 MCQ 2016 2M

A wire of length 340 mm is to be cut into two parts. One of the parts is to be made into a square and the other into a rectangle where sides are in the ratio 1:2. What is the length of the side of the square (in mm) such that the combined area of the square and the rectangle is a **MINIMUM**?

A. 30 **B.** 40 **C.** 120 **D.** 180

Session : 4 | Afternoon | 31/01/16

Q16.4.1 MCQ 2016 1M

An apple costs Rs. 10. An onion costs Rs. 8.

Select the most suitable sentence with respect to grammar and usage.

A. The price of an apple is greater than an onion.

B. The price of an apple is more than onion.

C. The price of an apple is greater than that of an onion.

D. Apples are more costlier than onions.

Q16.4.2 MCQ 2016 1M

The Buddha said, "Holding on to anger is like grasping a hot coal with the intent of throwing it at someone else; you are the one who gets burnt."

Select the word below which is closest in meaning to the word underlined above.

A. burning **B.** igniting **C.** clutching **D.** flinging

Q16.4.3 MCQ 2016 1M

M has a son **Q** and a daughter **R**. He has no other children. **E** is the mother of **P** and daughter-in-law of **M**. How is **P** related to **M**?

A. **P** is the son-in-law of **M**.

B. **P** is the grandchild of **M**.

C. **P** is the daughter-in law of **M**.

D. **P** is the grandfather of **M**.

Q16.4.4 MCQ 2016 1M

The number that least fits this set:
(324, 441, 97 and 64) is _____.

A. 324 **B.** 441 **C.** 97 **D.** 64

Q16.4.5 MCQ 2016 1M

It takes 10 s and 15 s, respectively, for two trains travelling at different constant speeds to completely pass a telegraph post. The length of the first train is 120 m and that of the second train is 150 m. The magnitude of the difference in the speeds of the two trains (in m/s) is _____.

A. 2.0 **B.** 10.0 **C.** 12.0 **D.** 22.0

Q16.4.6 MCQ 2016 2M

The velocity V of a vehicle along a straight line is measured in m/s and plotted as shown with respect to time in seconds. At the end of the 7 seconds, how much will the odometer reading increase by (in m)?

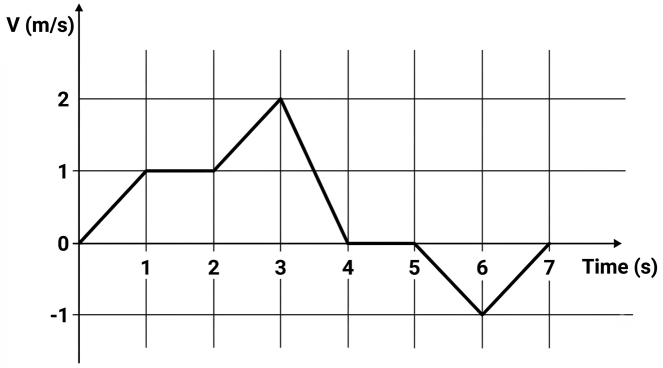


Fig. Q16.4.6

A. 0 **B.** 3 **C.** 4 **D.** 5

Q16.4.7 MCQ 2016 2M

The overwhelming number of people infected with rabies in India has been flagged by the World Health Organization as a source of concern. It is estimated that inoculating 70% of pets and stray dogs against rabies can lead to a significant reduction in the number of people infected with rabies.
Which of the following can be logically inferred from the above sentences?

A. The number of people in India infected with rabies is high.

B. The number of people in other parts of the world who are infected with rabies is low.

C. Rabies can be eradicated in India by vaccinating 70% of stray dogs.

D. Stray dogs are the main source of rabies worldwide.

Q16.4.8 MCQ 2016 2M

A flat is shared by four first year undergraduate students. They agreed to allow the oldest of them to enjoy some extra space in the flat. Manu is two months older than Sravan, who is three months younger than Trideep. Pavan is one month older than Sravan. Who should occupy the extra space in the flat?

A. Manu **B.** Sravan **C.** Trideep **D.** Pavan

Q16.4.9 MCQ 2016 2M

Find the area bounded by the lines $3x+2y = 14$, $2x-3y = 5$ in the first quadrant.

A. 14.95 **B.** 15.25 **C.** 15.70 **D.** 20.35

Q16.4.10 MCQ 2016 2M

A straight line is fit to a data set $(\ln x,\ y)$. This line intercepts the abscissa at $\ln x = 0.1$ and has a slope of -0.02. What is the value of y at $x = 5$ from the fit?

A. −0.030 **B.** −0.014 **C.** 0.014 **D.** 0.030

Session : 5 | Forenoon | 06/02/16

Q16.5.1 MCQ 2016 1M

Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

A. I will not leave the place until the minister does not meet me.

B. I will not leave the place until the minister doesn't meet me.

C. I will not leave the place until the minister meet me.

D. I will not leave the place until the minister meets me.

Q16.5.2 MCQ 2016 1M

A rewording of something written or spoken is a _____.

A. paraphrase **B.** paradox **C.** paradigm **D.** paraffin

Q16.5.3 MCQ 2016 1M

Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world." The sentence above is an example of a _____ statement.

A. figurative **B.** collateral **C.** literal **D.** figurine

Q16.5.4 MCQ 2016 1M

If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

A. zentaga **B.** tagafer **C.** tagazen **D.** relffer

Q16.5.5 MCQ 2016 1M

A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

A. 56 **B.** 64 **C.** 72 **D.** 96

Q16.5.6 MCQ 2016 2M

A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

A. Elegance **B.** Executive **C.** Smooth **D.** Soft

Q16.5.7 MCQ 2016 2M

Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.
Which of the following can be logically inferred from the above sentences?

A. India is a country of exactly seventeen languages.

B. Linguistic pluralism is the only indicator of a nation's diversity.

C. Indian currency notes have sufficient space for all the Indian languages.

D. Linguistic pluralism is strong evidence of India's diversity.

Q16.5.8 MCQ 2016 2M

Consider the following statements relating to the level of poker play of four players **P**, **Q**, **R** and **S**.

I. **P** always beats **Q**
II. **R** always beats **S**
III. **S** loses to **P** only sometimes
IV. **R** always loses to **Q**

Which of the following can be logically inferred from the above statements?

(i) **P** is likely to beat all the three other players
(ii) **S** is the absolute worst player in the set

A. (i) only **B.** (ii) only

C. (i) and (ii) **D.** neither (i) nor (ii)

Q16.5.9 MCQ 2016 2M

If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

A. $(x^3 + 8)$ **B.** $(x - 1)$ **C.** $(2x - 5)$ **D.** $(x + 1)$

Q16.5.10 MCQ 2016 2M

In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

A. 40.00 **B.** 46.02 **C.** 60.01 **D.** 92.02

Session : 6 | Afternoon | 06/02/16

Q16.6.1 MCQ 2016 1M

The man who is now Municipal Commissioner worked as _____.

A. the security guard at a university

B. a security guard at the university

C. a security guard at university

D. the security guard at the university

Q16.6.2 MCQ 2016 1M

Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.
Choose the option which is closest in meaning to the underlined phrase in the above sentence.

A. put up with **B.** put in with

C. put down to **D.** put up against

Q16.6.3 MCQ 2016 1M

Find the odd one in the following group of words.
mock, deride, praise, jeer

A. mock **B.** deride **C.** praise **D.** jeer

Q16.6.4 MCQ 2016 1M

Pick the odd one from the following options.

A. CADBE **B.** JHKIL **C.** XVYWZ **D.** ONPMQ

Q16.6.5 MCQ 2016 1M

In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

A. n^4 **B.** 4^n **C.** 2^{2n-1} **D.** 4^{n-1}

Q16.6.6 MCQ 2016 2M

Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

A. 35 **B.** 45 **C.** 65 **D.** 90

Q16.6.7 MCQ 2016 2M

Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.
Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

(i) The author believes that computers are not good for us.
(ii) Mobile computers and the internet are both intended inventions.

A. (i) only **B.** (ii) only

C. both (i) and (ii) **D.** neither (i) nor (ii)

Q16.6.8 MCQ 2016 2M

All hill-stations have a lake.
Ooty has two lakes.
Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

(i) Ooty is not a hill-station.
(ii) No hill-station can have more than one lake.

A. (i) only **B.** (ii) only

C. both (i) and (ii) **D.** neither (i) nor (ii)

Q16.6.9 MCQ 2016 2M

In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?



Fig. Q16.6.9

A. 21 **B.** 27 **C.** 30 **D.** 36

Q16.6.10 MCQ 2016 2M

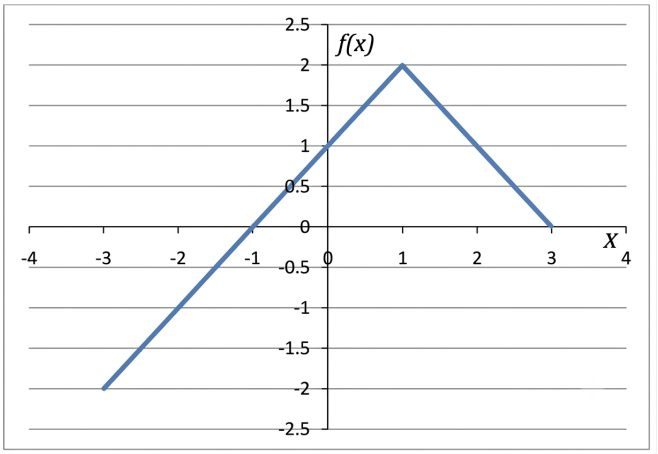


Fig. Q16.6.10

Choose the correct expression for $f(x)$ given in the graph.

A. $f(x) = 1 - |x - 1|$ **B.** $f(x) = 1 + |x - 1|$

C. $f(x) = 2 - |x - 1|$ **D.** $f(x) = 2 + |x - 1|$

Session : 7 | Forenoon | 07/02/16

Q16.7.1 MCQ 2016 1M

If I were you, I _____ that laptop. It's much too expensive.

A. won't buy **B.** shan't buy

C. wouldn't buy **D.** would buy

Q16.7.2 MCQ 2016 1M

He turned a deaf ear to my request.
What does the underlined phrasal verb mean?

A. ignored **B.** appreciated **C.** twisted **D.** returned

Q16.7.3 MCQ 2016 1M

Choose the most appropriate set of words from the options given below to complete the following sentence.
_____ _____ is a will, _____ is a way.

A. Wear, there, their **B.** Were, their, there

C. Where, there, there **D.** Where, their, their

Q16.7.4 MCQ 2016 1M

$(x\%$ of $y) + (y\%$ of $x)$ is equivalent to _____.

A. 2% of xy **B.** 2% of $(xy/100)$

C. $xy\%$ of 100 **D.** 100% of xy

Q16.7.5 MCQ 2016 1M

The sum of the digits of a two digit number is 12. If the new number formed by reversing the digits is greater than the original number by 54, find the original number.

A. 39 **B.** 57 **C.** 66 **D.** 93

Q16.7.6 MCQ 2016 2M

Two finance companies, P and Q, declared fixed annual rates of interest on the amounts invested with them. The rates of interest offered by these companies may differ from year to year. Year-wise annual rates of interest offered by these companies are shown by the line graph provided below.

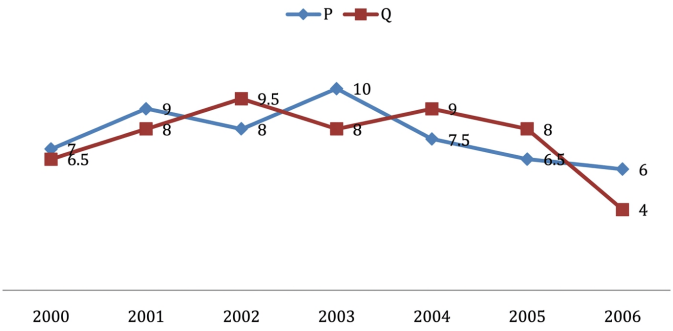


Fig. Q16.7.6

If the amounts invested in the companies, P and Q, in 2006 are in the ratio 8:9, then the amounts received after one year as interests from companies P and Q would be in the ratio:

A. 2:3 **B.** 3:4 **C.** 6:7 **D.** 4:3

Q16.7.7 MCQ 2016 2M

Today, we consider Ashoka as a great ruler because of the copious evidence he left behind in the form of stone carved edicts. Historians tend to correlate greatness of a king at his time with the availability of evidence today.
Which of the following can be logically inferred from the above sentences?

A. Emperors who do not leave significant sculpted evidence are completely forgotten.

B. Ashoka produced stone carved edicts to ensure that later historians will respect him.

C. Statues of kings are a reminder of their greatness.

D. A king's greatness, as we know him today, is interpreted by historians.

Q16.7.8 MCQ 2016 2M

Fact 1: Humans are mammals.
Fact 2: Some humans are engineers.
Fact 3: Engineers build houses.

If the above statements are facts, which of the following can be logically inferred?

I. All mammals build houses.
II. Engineers are mammals.
III. Some humans are not engineers.

A. II only **B.** III only

C. I, II and III **D.** I only

Q16.7.9 MCQ 2016 2M

A square pyramid has a base perimeter x, and the slant height is half of the perimeter. What is the lateral surface area of the pyramid?

A. x^2 **B.** $0.75\,x^2$ **C.** $0.50\,x^2$ **D.** $0.25\,x^2$

Q16.7.10 MCQ 2016 2M

Ananth takes 6 hours and Bharath takes 4 hours to read a book. Both started reading copies of the book at the same time. After how many hours is the number of pages **to be** read by Ananth, twice that **to be** read by Bharath? Assume Ananth and Bharath read all the pages with constant pace.

A. 1 **B.** 2 **C.** 3 **D.** 4

Session : 8 | Afternoon | 07/02/16

Q16.8.1 MCQ 2016 1M

The chairman requested the aggrieved shareholders to _____ him.

A. bare with **B.** bore with

C. bear with **D.** bare

Q16.8.2 MCQ 2016 1M

Identify the correct spelling out of the given options:

A. Managable **B.** Manageable

C. Mangaeble **D.** Managible

Q16.8.3 MCQ 2016 1M

Pick the odd one out in the following:
13, 23, 33, 43, 53

A. 23 **B.** 33 **C.** 43 **D.** 53

Q16.8.4 MCQ 2016 1M

R2D2 is a robot. R2D2 can repair aeroplanes. No other robot can repair aeroplanes.
Which of the following can be logically inferred from the above statements?

A. R2D2 is a robot which can only repair aeroplanes.

B. R2D2 is the only robot which can repair aeroplanes.

C. R2D2 is a robot which can repair only aeroplanes.

D. Only R2D2 is a robot.

Q16.8.5 MCQ 2016 1M

If $|9y - 6| = 3$, then $y^2 - 4y/3$ is _____.

A. 0 **B.** $+1/3$ **C.** $-1/3$ **D.** undefined

Q16.8.6 NAT 2016 2M

The following graph represents the installed capacity for cement production (in tonnes) and the actual production (in tonnes) of nine cement plants of a cement company. Capacity utilization of a plant is defined as ratio of actual production of cement to installed capacity. A plant with installed capacity of at least 200 tonnes is called a large plant and a plant with lesser capacity is called a small plant. The difference between total production of large plants and small plants, in tonnes is .

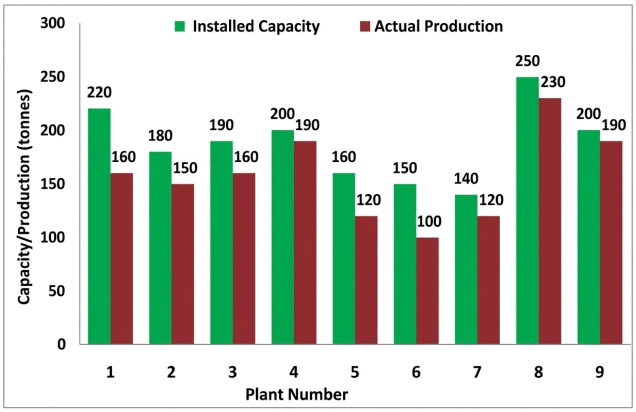


Fig. Q16.8.6

Q16.8.7 MCQ 2016 2M

A poll of students appearing for masters in engineering indicated that 60% of the students believed that mechanical engineering is a profession unsuitable for women. A research study on women with masters or higher degrees in mechanical engineering found that 99% of such women were successful in their professions.
Which of the following can be logically inferred from the above paragraph?

A. Many students have misconceptions regarding various engineering disciplines.

B. Men with advanced degrees in mechanical engineering believe women are well suited to be mechanical engineers.

C. Mechanical engineering is a profession well suited for women with masters or higher degrees in mechanical engineering.

D. The number of women pursuing higher degrees in mechanical engineering is small.

Q16.8.8 MCQ 2016 2M

Sourya committee had proposed the establishment of Sourya Institutes of Technology (SITs) in line with Indian Institutes of Technology (IITs) to cater to the technological and industrial needs of a developing country.
Which of the following can be logically inferred from the above sentence?

Based on the proposal,

(i) In the initial years, SIT students will get degrees from IIT.
(ii) SITs will have a distinct national objective.
(iii) SIT like institutions can only be established in consultation with IIT.
(iv) SITs will serve technological needs of a developing country.

A. (iii) and (iv) only. **B.** (i) and (iv) only.

C. (ii) and (iv) only. **D.** (ii) and (iii) only.

Q16.8.9 MCQ 2016 2M

Shaquille O' Neal is a 60% career free throw shooter, meaning that he successfully makes 60 free throws out of 100 attempts on average. What is the probability that he will successfully make exactly 6 free throws in 10 attempts?

A. 0.2508 **B.** 0.2816 **C.** 0.2934 **D.** 0.6000

Q16.8.10 NAT 2016 2M

The numeral in the units position of $211^{870} + 146^{127} \times 3^{424}$ is _____.

Answer Key

Session : 1					
Q16.1.1 A	Q16.1.2 B	Q16.1.3 A	Q16.1.4 A	Q16.1.5 C	Q16.1.6 C
Q16.1.7 B	Q16.1.8 D	Q16.1.9 C	Q16.1.10 C		
Session : 2					
Q16.2.1 B	Q16.2.2 A	Q16.2.3 D	Q16.2.4 C	Q16.2.5 B	Q16.2.6 C
Q16.2.7 C	Q16.2.8 C	Q16.2.9 A	Q16.2.10 C		
Session : 3					
Q16.3.1 C	Q16.3.2 B	Q16.3.3 C	Q16.3.4 C	Q16.3.5 B	Q16.3.6 D
Q16.3.7 A	Q16.3.8 D	Q16.3.9 C	Q16.3.10 B		
Session : 4					
Q16.4.1 C	Q16.4.2 C	Q16.4.3 B	Q16.4.4 C or D	Q16.4.5 A	Q16.4.6 D
Q16.4.7 A	Q16.4.8 C	Q16.4.9 B	Q16.4.10 A		
Session : 5					
Q16.5.1 D	Q16.5.2 A	Q16.5.3 A	Q16.5.4 C	Q16.5.5 D	Q16.5.6 B
Q16.5.7 D	Q16.5.8 D	Q16.5.9 B	Q16.5.10 B		
Session : 6					
Q16.6.1 B	Q16.6.2 A	Q16.6.3 C	Q16.6.4 D	Q16.6.5 B	Q16.6.6 A
Q16.6.7 D	Q16.6.8 D	Q16.6.9 C	Q16.6.10 C		
Session : 7					
Q16.7.1 C	Q16.7.2 A	Q16.7.3 C	Q16.7.4 A	Q16.7.5 A	Q16.7.6 D
Q16.7.7 D	Q16.7.8 B	Q16.7.9 D	Q16.7.10 C		
Session : 8					
Q16.8.1 C	Q16.8.2 B	Q16.8.3 B	Q16.8.4 B	Q16.8.5 C	Q16.8.6 120.00
Q16.8.7 C	Q16.8.8 C	Q16.8.9 A	Q16.8.10 7.00		

YEAR

2015

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AE, AR, BT, CH, CY, EC-1, EY, XE, XL	10
Session 2	EC-2, ME-1	10
Session 3	EC-3, ME-2	10
Session 4	AE, GG, IN, MA, ME-3, MN, MT, PH, PI, TF	10
Session 5	CS-1, EE-1	10
Session 6	CS-2, EE-2	10
Session 7	CE-1, CS-3	10
Session 8	CE-2	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 72 **MSQ:** 0 **NAT:** 8

Session : 1 | Forenoon | 31/01/15

Q15.1.1 MCQ 2015 1M

Choose the most appropriate word from the options given below to complete the following sentence.
The principal presented the chief guest with a _____, as token of appreciation.

A. momento **B.** memento **C.** momentum **D.** moment

Q15.1.2 MCQ 2015 1M

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:
Frogs _____.

A. croak **B.** roar **C.** hiss **D.** patter

Q15.1.3 MCQ 2015 1M

Choose the word most similar in meaning to the given word:
Educe

A. Exert **B.** Educate **C.** Extract **D.** Extend

Q15.1.4 MCQ 2015 1M

Operators $\square$, $\diamond$ and $\rightarrow$ are defined by:

$$a\square b = \frac{a-b}{a+b}; a\diamond b = \frac{a+b}{a-b}; a \rightarrow b = ab.$$

Find the value of $(66\square 6) \rightarrow (66\diamond 6)$.

A. -2 **B.** -1 **C.** 1 **D.** 2

Q15.1.5 MCQ 2015 1M

If $\log_x(5/7) = -1/3$, then the value of x is

A. 343/125 **B.** 125/343 **C.** -25/49 **D.** -49/25

Q15.1.6 MCQ 2015 2M

The following question presents a sentence, part of which is underlined. Beneath the sentence you find four ways of phrasing the underlined part. Following the requirements of the standard written English, select the answer that produces the most effective sentence.
Tuberculosis, together with its effects, ranks one of the leading causes of death in India.

A. ranks as one of the leading causes of death

B. rank as one of the leading causes of death

C. has the rank of one of the leading causes of death

D. are one of the leading causes of death

Q15.1.7 MCQ 2015 2M

Read the following paragraph and choose the correct statement.
Climate change has reduced human security and threatened human well being. An ignored reality of human progress is that human security largely depends upon environmental security. But on the contrary, human progress seems contradictory to environmental security. To keep up both at the required level is a challenge to be addressed by one and all. One of the ways to curb the climate change may be suitable scientific innovations, while the other may be the Gandhian perspective on small scale progress with focus on sustainability.

A. Human progress and security are positively associated with environmental security.

B. Human progress is contradictory to environmental security.

C. Human security is contradictory to environmental security.

D. Human progress depends upon environmental security.

Q15.1.8 NAT 2015 2M

Fill in the missing value

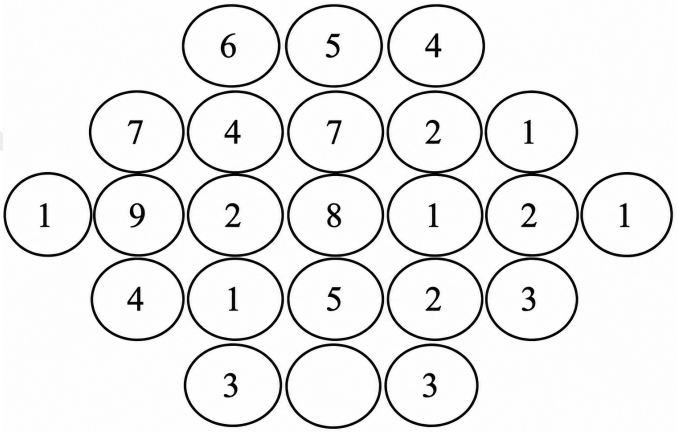


Fig. Q15.1.8

Q15.1.9 MCQ 2015 2M

A cube of side 3 units is formed using a set of smaller cubes of side 1 unit. Find the proportion of the number of faces of the smaller cubes visible to those which are NOT visible.

A. 1 : 4 **B.** 1 : 3 **C.** 1 : 2 **D.** 2 : 3

Q15.1.10 MCQ 2015 2M

Humpty Dumpty sits on a wall every day while having lunch. The wall sometimes breaks. A person sitting on the wall falls if the wall breaks.
Which one of the statements below is logically valid and can be inferred from the above sentences?

A. Humpty Dumpty always falls while having lunch

B. Humpty Dumpty does not fall sometimes while having lunch

C. Humpty Dumpty never falls during dinner

D. When Humpty Dumpty does not sit on the wall, the wall does not break

Session : 2 | Afternoon | 31/01/15

Q15.2.1 MCQ 2015 1M

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:
Dhoni, as well as the other team members of Indian team, _____ present on the occasion.

A. were **B.** was **C.** has **D.** have

Q15.2.2 MCQ 2015 1M

Choose the word most similar in meaning to the given word:

Awkward

A. Inept **B.** Graceful **C.** Suitable **D.** Dreadful

Q15.2.3 MCQ 2015 1M

What is the adverb for the given word below?

Misogynous

A. Misogynousness **B.** Misogynity

C. Misogynously **D.** Misogynous

Q15.2.4 MCQ 2015 1M

An electric bus has onboard instruments that report the total electricity consumed since the start of the trip as well as the total distance covered. During a single day of operation, the bus travels on stretches M, N, O, and P, in that order. The cumulative distances travelled and the corresponding electricity consumption are shown in the Table below:

Stretch	Cumulative distance (km)	Electricity used (kWh)
M	20	12
N	45	25
O	75	45
P	100	57

The stretch where the electricity consumption per km is minimum is

A. M **B.** N **C.** O **D.** P

Q15.2.5 MCQ 2015 1M

Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is $\frac{1}{6}$ and that of Ramesh is $\frac{1}{8}$. What is the probability that only one of them will be selected?

A. $\frac{47}{48}$ **B.** $\frac{1}{4}$ **C.** $\frac{13}{48}$ **D.** $\frac{35}{48}$

Q15.2.6 MCQ 2015 2M

In the following sentence certain parts are underlined and marked P, Q, and R. One of the parts may contain certain error or may not be acceptable in standard written communication. Select the part containing an error. Choose D as your answer if there is no error.
The student corrected all the errors (P) that the instructor marked (Q) on the answer book (R).

A. P **B.** Q **C.** R **D.** No Error

Q15.2.7 MCQ 2015 2M

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:

I. All film stars are playback singers.
II. All film directors are film stars.

Conclusions:

I. All film directors are playback singers.
II. Some film stars are film directors.

A. Only conclusion I follows.

B. Only conclusion II follows.

C. Neither conclusion I nor II follows.

D. Both conclusions I and II follow.

Q15.2.8 NAT 2015 2M

A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 metre and 5 metre per leap respectively, what distance in metres will the tiger have to run before it catches the deer?

Q15.2.9 MCQ 2015 2M

If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ac$ lies in the interval

A. $[1, \frac{2}{3}]$ **B.** $[-\frac{1}{2}, 1]$ **C.** $[-1, \frac{1}{2}]$ **D.** $[2, -4]$

Q15.2.10 MCQ 2015 2M

Lamenting the gradual sidelining of the arts in school curricula, a group of prominent artists wrote to the Chief Minister last year, asking him to allocate more funds to support arts education in schools. However, no such increase has been announced in this year's Budget. The artists expressed their deep anguish at their request not being approved, but many of them remain optimistic about funding in the future.

Which of the statement(s) below is/are logically valid and can be inferred from the above statements?

(i) The artists expected funding for the arts to increase this year.
(ii) The Chief Minister was receptive to the idea of increasing funding for the arts.
(iii) The Chief Minister is a prominent artist.
(iv) Schools are giving less importance to arts education nowadays.

A. (iii) and (iv) **B.** (i) and (iv)

C. (i), (ii) and (iv) **D.** (i) and (iii)

Session : 3 | Forenoon | 01/02/15

Q15.3.1 MCQ 2015 1M

Choose the most appropriate word from the options given below to complete the following sentence.

If the athlete had wanted to come first in the race, he _____ several hours every day.

A. should practise **B.** should have practised

C. practised **D.** should be practising

Q15.3.2 MCQ 2015 1M

Choose the most suitable one word substitute for the following expression:

Connotation of a road or way

A. Pertinacious **B.** Viaticum

C. Clandestine **D.** Ravenous

Q15.3.3 MCQ 2015 1M

Choose the correct verb to fill in the blank below:

Let us _____.

A. introvert **B.** alternate **C.** atheist **D.** altruist

Q15.3.4 MCQ 2015 1M

Find the missing sequence in the letter series below:

A, CD, GHI, ?, UVWXY

A. LMN **B.** MNO **C.** MNOP **D.** NOPQ

Q15.3.5 MCQ 2015 1M

If $x > y > 1$, which of the following must be true?

i. $\ln x > \ln y$ iii. $y^x > x^y$

ii. $e^x > e^y$ iv. $\cos x > \cos y$

A. (i) and (ii) **B.** (i) and (iii)

C. (iii) and (iv) **D.** (ii) and (iv)

Q15.3.6 MCQ 2015 2M

Ram and Shyam shared a secret and promised to each other that it would remain between them. Ram expressed himself in one of the following ways as given in the choices below. Identify the correct way as per standard English.

A. It would remain between you and me.

B. It would remain between I and you.

C. It would remain between you and I.

D. It would remain with me.

Q15.3.7 MCQ 2015 2M

In the following question, the first and the last sentence of the passage are in order and numbered 1 and 6. The rest of the passage is split into 4 parts and numbered as 2, 3, 4, and 5. These 4 parts are not arranged in proper order. Read the sentences and arrange them in a logical sequence to make a passage and choose the correct sequence from the given options.

1. On Diwali, the family rises early in the morning.
2. The whole family, including the young and the old enjoy doing this.
3. Children let off fireworks later in the night with their friends.
4. At sunset, the lamps are lit and the family performs various rituals.

5. Father, mother, and children visit relatives and exchange gifts and sweets.
6. Houses look so pretty with lighted lamps all around.

A. 2,5,3,4 **B.** 5,2,4,3 **C.** 3,5,4,2 **D.** 4,5,2,3

Q15.3.8 NAT 2015 2M

From a circular sheet of paper of radius 30 cm, a sector of 10% area is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is _____.

Q15.3.9 MCQ 2015 2M

$\log \tan 1^\circ + \log \tan 2^\circ + \ldots\ldots + \log \tan 89^\circ$ is. . .

A. 1 **B.** $\frac{1}{\sqrt{2}}$ **C.** 0 **D.** -1

Q15.3.10 MCQ 2015 2M

Ms. X will be in Bagdogra from 01/05/2014 to 20/05/2014 and from 22/05/2014 to 31/05/2014. On the morning of 21/05/2014, she will reach Kochi via Mumbai.
Which one of the statements below is logically valid and can be inferred from the above sentences?

A. Ms. X will be in Kochi for one day, only in May.

B. Ms. X will be in Kochi for only one day in May.

C. Ms. X will be only in Kochi for one day in May.

D. Only Ms. X will be in Kochi for one day in May.

Session : 4 | Afternoon | 01/02/15

Q15.4.1 MCQ 2015 1M

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:

Apparent lifelessness _____ dormant life.

A. harbours **B.** leads to **C.** supports **D.** affects

Q15.4.2 MCQ 2015 1M

Fill in the blank with the correct idiom/phrase.

That boy from the town was a _____ in the sleepy village.

A. dog out of herd **B.** sheep from the heap

C. fish out of water **D.** bird from the flock

Q15.4.3 MCQ 2015 1M

Choose the statement where underlined word is used correctly.

A. When the teacher eludes to different authors, he is being elusive.

B. When the thief keeps eluding the police, he is being elusive.

C. Matters that are difficult to understand, identify or remember are allusive.

D. Mirages can be allusive, but a better way to express them is illusory.

Q15.4.4 MCQ 2015 1M

Tanya is older than Eric.
Cliff is older than Tanya.
Eric is older than Cliff.

If the first two statements are true, then the third statement is:

A. True **B.** False **C.** Uncertain **D.** Data insufficient

Q15.4.5 MCQ 2015 1M

Five teams have to compete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have to be held to complete the league round of matches?

A. 20 **B.** 10 **C.** 8 **D.** 5

Q15.4.6 MCQ 2015 2M

Select the appropriate option in place of underlined part of the sentence.
Increased productivity necessary reflects greater efforts made by the employees.

A. Increase in productivity necessary

B. Increase productivity is necessary

C. Increase in productivity necessarily

D. No improvement required

Q15.4.7 MCQ 2015 2M

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:

I. No manager is a leader.
II. All leaders are executives.

Conclusions:

I. No manager is an executive.

II. No executive is a manager.

A. Only conclusion I follows.

B. Only conclusion II follows.

C. Neither conclusion I nor II follows.

D. Both conclusions I and II follow.

Q15.4.8 NAT 2015 2M

In the given figure angle Q is a right angle, $PS : QS = 3 : 1$, $RT : QT = 5 : 2$ and $PU : UR = 1 : 1$. If area of triangle QTS is 20 cm^2, then the area of triangle PQR in cm^2 is _____.

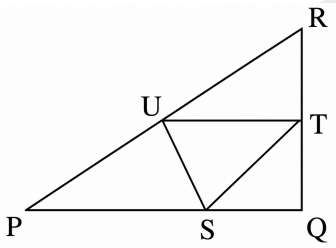


Fig. Q15.4.8

Q15.4.9 MCQ 2015 2M

Right triangle PQR is to be constructed in the xy-plane so that the right angle is at P and line PR is parallel to the x-axis. The x and y coordinates of P, Q, and R are to be integers that satisfy the inequalities: $-4 \leq x \leq 5$ and $6 \leq y \leq 16$. How many different triangles could be constructed with these properties?

A. 110 **B.** 1,100 **C.** 9,900 **D.** 10,000

Q15.4.10 MCQ 2015 2M

A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statements is TRUE?

A. X and Y are not independent

B. Y and Z are dependent

C. Y and Z are independent

D. X and Z are independent

Session : 5 | Forenoon | 07/02/15

Q15.5.1 MCQ 2015 1M

Didn't you buy _____ when you went shopping?

A. any paper **B.** much paper

C. no paper **D.** a few paper

Q15.5.2 MCQ 2015 1M

Which of the following options is the closest in meaning to the sentence below?

She enjoyed herself immensely at the party.

A. She had a terrible time at the party

B. She had a horrible time at the party

C. She had a terrific time at the party

D. She had a terrifying time at the party

Q15.5.3 MCQ 2015 1M

Which one of the following combinations is incorrect?

A. Acquiescence – Submission

B. Wheedle – Roundabout

C. Flippancy – Lightness

D. Profligate – Extravagant

Q15.5.4 MCQ 2015 1M

Based on the given statements, select the most appropriate option to solve the given question.

If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building?

Statements:

I. Each step is 3/4 foot high.

II. Each step is 1 foot wide.

A. Statement I alone is sufficient, but statement II alone is not sufficient.

B. Statement II alone is sufficient, but statement I alone is not sufficient.

C. Both statements are sufficient, but neither statement alone is sufficient.

D. Statement I and II together are not sufficient.

Q15.5.5 MCQ 2015 1M

Given Set $A = \{2, 3, 4, 5\}$ and Set $B = \{11, 12, 13, 14, 15\}$, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

A. 0.20 **B.** 0.25 **C.** 0.30 **D.** 0.33

Q15.5.6 MCQ 2015 2M

Select the alternative meaning of the underlined part of the sentence.

The chain snatchers took to their heels when the police party arrived.

A. took shelter in a thick jungle

B. open indiscriminate fire

C. took to flight

D. unconditionally surrendered

Q15.5.7 MCQ 2015 2M

The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option.

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

I. The water supply authority should impose a partial cut in supply to tackle the situation.

II. The government should appeal to all the residents through mass media for minimal use of water.

III. The government should ban the water supply in lower areas.

A. Statements I and II follow.

B. Statements I and III follow.

C. Statements II and III follow.

D. All statements follow.

Q15.5.8 NAT 2015 2M

The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5:4. There are 40 males in Electrical Engineering. What is the difference between the numbers of female students in the Civil department and the female students in the Mechanical department?

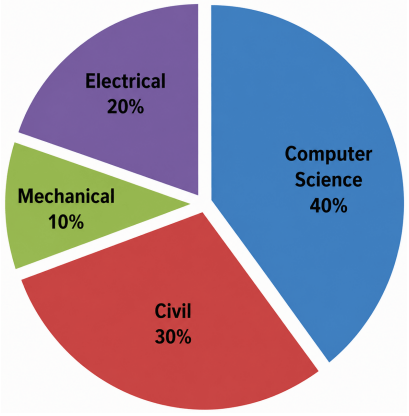


Fig. Q15.5.8

Q15.5.9 MCQ 2015 2M

The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c:

I. $p + m + c = 27/20$

II. $p + m + c = 13/20$

III. $(p) \cdot (m) \cdot (c) = 1/10$

A. Only relation I is true.

B. Only relation II is true.

C. Relations II and III are true.

D. Relations I and III are true.

Q15.5.10 MCQ 2015 2M

The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

A. 2.290 **B.** 2.970 **C.** 6.795 **D.** 8.795

Session : 6 | Afternoon | 07/02/15

Q15.6.1 MCQ 2015 1M

We _____ our friend's birthday and we _____ how to make it up to him.

A. completely forgot — don't just know

B. forgot completely — don't just know

C. completely forgot — just don't know

D. forgot completely — just don't know

Q15.6.2 MCQ 2015 1M

Choose the statement where underlined word is used correctly.

A. The industrialist had a personnel jet.

B. I write my experience in my personnel diary.

C. All personnel are being given the day off.

D. Being religious is a personnel aspect.

Q15.6.3 MCQ 2015 1M

A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is

A. fabric **B.** textile **C.** fibre **D.** apparel

Q15.6.4 MCQ 2015 1M

Based on the given statements, select the most appropriate option to solve the given question.

What will be the total weight of 10 poles each of same weight?

Statements:

I. One fourth of the weight of a pole is 5 Kg.

II. The total weight of these poles is 160 kg more than the total weight of two poles.

A. Statement I alone is not sufficient.

B. Statement II alone is not sufficient.

C. Either I or II alone is sufficient.

D. Both statements I and II together are not sufficient.

Q15.6.5 MCQ 2015 1M

Consider a function $f(x) = 1 - |x|$ on $-1 \le x \le 1$. The value of x at which the function attains a maximum, and the maximum value of the function are:

A. $0, -1$ **B.** $-1, 0$ **C.** $0, 1$ **D.** $-1, 2$

Q15.6.6 MCQ 2015 2M

Out of the following four sentences, select the most suitable sentence with respect to grammar and usage:

A. Since the report lacked needed information, it was of no use to them.

B. The report was useless to them because there were no needed information in it.

C. Since the report did not contain the needed information, it was not real useful to them.

D. Since the report lacked needed information, it would not had been useful to them.

Q15.6.7 MCQ 2015 2M

In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60°$. What is the length of PS?

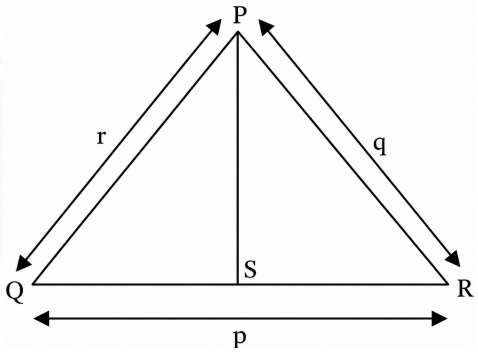


Fig. Q15.6.7

A. $\frac{(q+r)}{qr}$ **B.** $\frac{qr}{(q+r)}$

C. $\sqrt{(q^2+r^2)}$ **D.** $\frac{(q+r)^2}{qr}$

Q15.6.8 NAT 2015 2M

For distinct integers p, q, r, s, define the functions f, g, and h as follows:

$$f(p,q,r,s) = \max(p,q,r,s).$$

$$g(p,q,r,s) = \min(p,q,r,s).$$

The function $h(p,q,r,s)$ is the remainder when the larger of the two products pq and rs is divided by the smaller (for the unequal nonzero products used here). Products of the function values on the same arguments are denoted by $fg = f \times g$ and $fgh = f \times g \times h$. The same maximum/minimum operations also apply to two arguments. What is the value of $fg(h(2,5,7,3),4,6,8)$?

Q15.6.9 MCQ 2015 2M

If the list of letters, P, R, S, T, U is an arithmetic sequence, which of the following are also in arithmetic sequence?

I. $2P, 2R, 2S, 2T, 2U$
II. $P-3, R-3, S-3, T-3, U-3$
III. P^2, R^2, S^2, T^2, U^2

A. I only **B.** I and II **C.** II and III **D.** I and III

Q15.6.10 MCQ 2015 2M

Four branches of a company are located at M, N, O, and P. M is north of N at a distance of 4 km; P is south of O at a distance of 2 km; N is southeast of O by 1 km. What is the distance between M and P in km?

A. 5.34 **B.** 6.74 **C.** 28.5 **D.** 45.49

Session : 7 | Forenoon | 08/02/15

Q15.7.1 MCQ 2015 1M

Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian students that they close their minds to anything _____ to the requirements of the exam.

A. related **B.** extraneous **C.** outside **D.** useful

Q15.7.2 MCQ 2015 1M

Select the pair that best expresses a relationship similar to that expressed in the pair:

Children : Pediatrician

A. Adult : Orthopaedist **B.** Females : Gynaecologist

C. Kidney : Nephrologist **D.** Skin : Dermatologist

Q15.7.3 MCQ 2015 1M

The Tamil version of ____ John Abraham-starrer *Madras Cafe* ____ cleared by the Censor Board with no cuts last week, but the film's distributors _____ no takers among the exhibitors for a release in Tamil Nadu ____ this Friday.

A. Mr., was, found, on **B.** a, was, found, at

C. the, was, found, on **D.** a, being, find at

Q15.7.4 MCQ 2015 1M

If ROAD is written as URDG, then SWAN should be written as:

A. VXDQ **B.** VZDQ **C.** VZDP **D.** UXDQ

Q15.7.5 MCQ 2015 1M

A function $f(x)$ is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$.

A. 59 **B.** 45 **C.** 43 **D.** 35

Q15.7.6 MCQ 2015 2M

Alexander turned his attention towards India, since he had conquered Persia.
Which one of the statements below is logically valid and can be inferred from the above sentence?

A. Alexander would not have turned his attention towards India had he not conquered Persia.

B. Alexander was not ready to rest on his laurels, and wanted to march to India.

C. Alexander was completely in control of his army and could command it to move towards India.

D. Since Alexander's kingdoms extended to Indian borders after the conquest of Persia, he was keen to move further.

Q15.7.7 MCQ 2015 2M

Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order, he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several times after working hard to lay a strong foundation. His critics pointed out that until he addressed this problem, success at the highest level will continue to elude him.
Which of the statement(s) below is/are logically valid and can be inferred from the above passage?

(i) He was already a successful batsman at the highest level.
(ii) He has to improve his temperament in order to become a great batsman.
(iii) He failed to make many of his good starts count.
(iv) Improving his technical skills will guarantee success.

A. (iii) and (iv) **B.** (ii) and (iii)

C. (i), (ii) and (iii) **D.** (iv) only

Q15.7.8 NAT 2015 2M

The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?

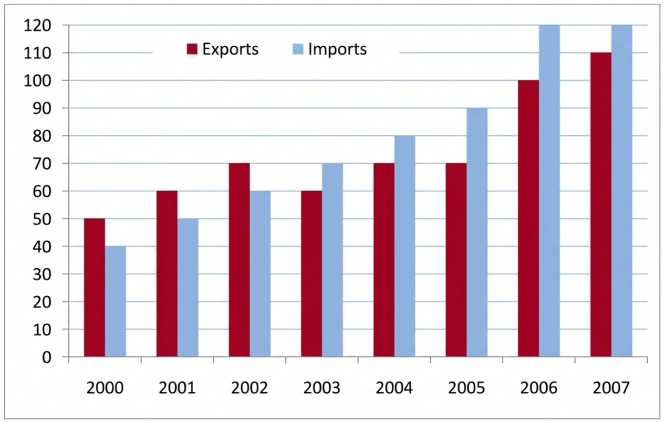


Fig. Q15.7.8

Q15.7.9 MCQ | 2015 | 2M

Choose the most appropriate equation for the function drawn as a thick line, in the plot below.

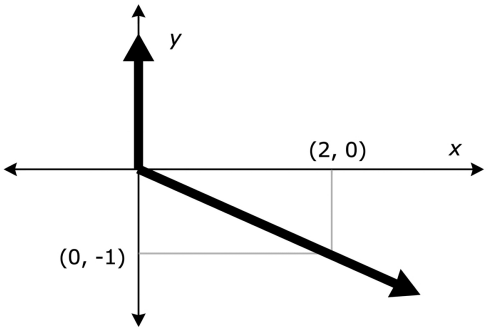


Fig. Q15.7.9

A. $x = y - |y|$ **B.** $x = -(y - |y|)$

C. $x = y + |y|$ **D.** $x = -(y + |y|)$

Q15.7.10 MCQ | 2015 | 2M

The head of a newly formed government desires to appoint five of the six selected members P, Q, R, S, T, and U to portfolios of Home, Power, Defense, Telecom, and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets either Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.
Which is the valid distribution of portfolios?

A. P-Home, Q-Power, R-Defense, S-Telecom, T-Finance

B. R-Home, S-Power, P-Defense, Q-Telecom, T-Finance

C. P-Home, Q-Power, T-Defense, S-Telecom, U-Finance

D. Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

Session : 8 | Afternoon | 08/02/15

Q15.8.1 MCQ | 2015 | 1M

Choose the most appropriate word from the options given below to complete the following sentence.
The official answered _____ that the complaints of the citizen would be looked into.

A. respectably **B.** respectfully

C. reputably **D.** respectively

Q15.8.2 MCQ | 2015 | 1M

Choose the statement where underlined word is used correctly.

A. The minister insured the victims that everything would be all right.

B. He ensured that the company will not have to bear any loss.

C. The actor got himself ensured against any accident.

D. The teacher insured students of good results.

Q15.8.3 MCQ | 2015 | 1M

Which word is not a synonym for the word *vernacular*?

A. regional **B.** indigenous **C.** indigent **D.** colloquial

Q15.8.4 MCQ | 2015 | 1M

Mr. Vivek walks 6 meters North-east, then turns and walks 6 meters South-east, both at 60 degrees to east. He further moves 2 meters South and 4 meters West. What is the straight distance in metres between the point he started from and the point he finally reached?

A. $2\sqrt{2}$ **B.** 2 **C.** $\sqrt{2}$ **D.** $1/\sqrt{2}$

Q15.8.5 MCQ | 2015 | 1M

Four cards are randomly selected from a pack of 52 cards. If the first two cards are kings, what is the probability that the third card is a king?

A. $4/52$ **B.** $2/50$

C. $(1/52) \times (1/52)$ **D.** $(1/52) \times (1/51) \times (1/50)$

Q15.8.6 MCQ | 2015 | 2M

The word similar in meaning to 'dreary' is

A. cheerful **B.** dreamy **C.** hard **D.** dismal

Q15.8.7 MCQ 2015 2M

The given question is followed by two statements; select the most appropriate option that solves the question.
Capacity of a solution tank A is 70% of the capacity of tank B. How many gallons of solution are in tank A and tank B?

Statements:

I. Tank A is 80% full and tank B is 40% full.
II. Tank A if full contains 14,000 gallons of solution.

A. Statement I alone is sufficient.

B. Statement II alone is sufficient.

C. Either statement I or II alone is sufficient.

D. Both the statements I and II together are sufficient.

Q15.8.8 NAT 2015 2M

How many four digit numbers can be formed with the 10 digits $0, 1, 2, \ldots, 9$ if no number can start with 0 and if repetitions are not allowed?

Q15.8.9 MCQ 2015 2M

Read the following table giving sales data of five types of batteries for years 2006 to 2012:

Year	Type I	Type II	Type III	Type IV	Type V
2006	75	144	114	102	108
2007	90	126	102	84	126
2008	96	114	75	105	135
2009	105	90	150	90	75
2010	90	75	135	75	90
2011	105	60	165	45	120
2012	115	85	160	100	145

Out of the following, which type of battery achieved highest growth between the years 2006 and 2012?

A. Type V **B.** Type III

C. Type II **D.** Type I

Q15.8.10 MCQ 2015 2M

There are 16 teachers who can teach Thermodynamics (TD), 11 who can teach Electrical Sciences (ES), and 5 who can teach both TD and Engineering Mechanics (EM). There are a total of 40 teachers. 6 cannot teach any of the three subjects, i.e. EM, ES or TD. 6 can teach only ES. 4 can teach all three subjects, i.e. EM, ES and TD. 4 can teach ES and TD. How many can teach both ES and EM but not TD?

A. 1 **B.** 2 **C.** 3 **D.** 4

Answer Key

Session : 1					
Q15.1.1 B	Q15.1.2 A	Q15.1.3 C	Q15.1.4 C	Q15.1.5 A	Q15.1.6 A
Q15.1.7 B	Q15.1.8 3	Q15.1.9 C	Q15.1.10 B		
Session : 2					
Q15.2.1 B	Q15.2.2 A	Q15.2.3 C	Q15.2.4 D	Q15.2.5 B	Q15.2.6 B
Q15.2.7 D	Q15.2.8 800.00	Q15.2.9 B	Q15.2.10 B		
Session : 3					
Q15.3.1 B	Q15.3.2 B	Q15.3.3 B	Q15.3.4 C	Q15.3.5 A	Q15.3.6 A
Q15.3.7 B	Q15.3.8 1.9 to 2.2	Q15.3.9 C	Q15.3.10 B		
Session : 4					
Q15.4.1 A	Q15.4.2 C	Q15.4.3 B	Q15.4.4 B	Q15.4.5 B	Q15.4.6 C
Q15.4.7 C	Q15.4.8 280.00	Q15.4.9 C	Q15.4.10 B		
Session : 5					
Q15.5.1 A	Q15.5.2 C	Q15.5.3 B	Q15.5.4 A	Q15.5.5 A	Q15.5.6 C
Q15.5.7 A	Q15.5.8 32.00	Q15.5.9 D	Q15.5.10 C		
Session : 6					
Q15.6.1 C	Q15.6.2 C	Q15.6.3 D	Q15.6.4 C	Q15.6.5 C	Q15.6.6 A
Q15.6.7 B	Q15.6.8 8.00	Q15.6.9 B	Q15.6.10 A		
Session : 7					
Q15.7.1 B	Q15.7.2 B	Q15.7.3 C	Q15.7.4 B	Q15.7.5 C	Q15.7.6 A
Q15.7.7 B	Q15.7.8 2006.00	Q15.7.9 B	Q15.7.10 B		
Session : 8					
Q15.8.1 B	Q15.8.2 B	Q15.8.3 C	Q15.8.4 A	Q15.8.5 B	Q15.8.6 D
Q15.8.7 D	Q15.8.8 4536	Q15.8.9 D	Q15.8.10 A		

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NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

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